

Value of Capacity Flexibility in Manufacturing Systems with Seasonal Demands

Sanket Bhat and Ananth Krishnamurthy

Department of Industrial and Systems Engineering, University of Wisconsin–Madison, USA

Abstract

In this paper, we analyze a manufacturing system subject to seasonal demands. The system is modeled as a single stage production facility with flexible production rate, and seasonal demands are modeled using a Markov modulated Poisson process. Using dynamic programming, we develop optimal policies under the infinite horizon discounted expected cost and the infinite horizon average expected cost criteria. We prove that the optimal policy is a season dependent base-stock policy with state-dependent production rates. Further, we identify the monotonic structure of the optimal policy with respect to the net inventory, the demand in a season, and the overall workload on the system. We, then, illustrate the value of joint flexibility in capacity and inventory levels under the seasonal demand settings. Numerical comparisons with the policies used in practice demonstrate the value of capacity flexibility.

Keywords: Production control, Flexible capacity, Seasonal demands, Markov Decision Process

1 Introduction

This paper investigates production and inventory control policies for manufacturing firms that experience seasonal demands for their products. Our research is motivated by collaborations with several manufacturing industries, including manufacturers of farm equipment (John Deere), motor cycles (Harley-Davidson), and theater lighting systems (Electronic Theatre Controls). Our collaborations suggest that manufacturers experience seasonal demands not just at the annual time scale (where demands show particular trends for periods ranging from 2 to 4 months), but also on shorter time scales that range in terms of 2 to 4 weeks. These ‘seasonal’ trends could be due to

factors such as market conditions, sales initiatives, seasonal behavior of customers, model changes or other consumer preferences. Further, in many cases, these demand variations could be significant: the demand during a peak season could be twice the demand observed in a low season. Most manufacturers respond to these variations by investing in flexibility in (i) inventory stocking levels, or (ii) capacity levels, or (iii) both. However, determining the optimal strategies to deal with various aspects of seasonality is a hard problem. In this research, we provide insights related to the value of investments in both capacity and inventory flexibility through analysis at two levels. First, we develop an analytical model that is able to capture the various attributes of seasonality in demands. We use this model to investigate which attributes of seasonality have a significant impact on manufacturing costs. Then, we use the model to investigate the structure and properties of the optimal policy for a system with seasonal demands.

To understand the manufacturing context, consider a typical demand and workload scenario experienced by John Deere’s stamping facility in Horicon, Wis. The facility manufactures components for farm and agricultural equipment such as tractors, harvesters, and lawn mowers and delivers these components to the assembly facility located nearby. Demands for these end products from retailers directly translate into demands on the assembly lines. These demands in turn impact the workload at the various workstations in the manufacturing facility. Figure 1 depicts the demand and workload at two workstations in this facility. For workstation A, in months of September to November, the workload is under two shifts, while in the months of February, March and April, the workload exceeds 3 shifts. In other months the workload is in between. At workstation B, the workload profile has a slightly different pattern, with low demands between September and December and high demands in June and July. Although the demand patterns are different, it can be argued that the demand at the workstations falls into three distinct ‘seasons’ (low demand, medium demand, and high demands) with each ‘season’ being roughly 60 to 120 days long. Consider next, the demand and workload profiles for the other workstations shown in Figure 2. At workstation C, the weekly workload and demand exceeds 3 shifts in weeks 2, 5 and 8 while at workstation D, the workload and demands are less than 2 shifts in weeks 1 through 4 and week 9. Again, it can be argued that demands demonstrate similar trends, although at a weekly time scale (5-10 days).

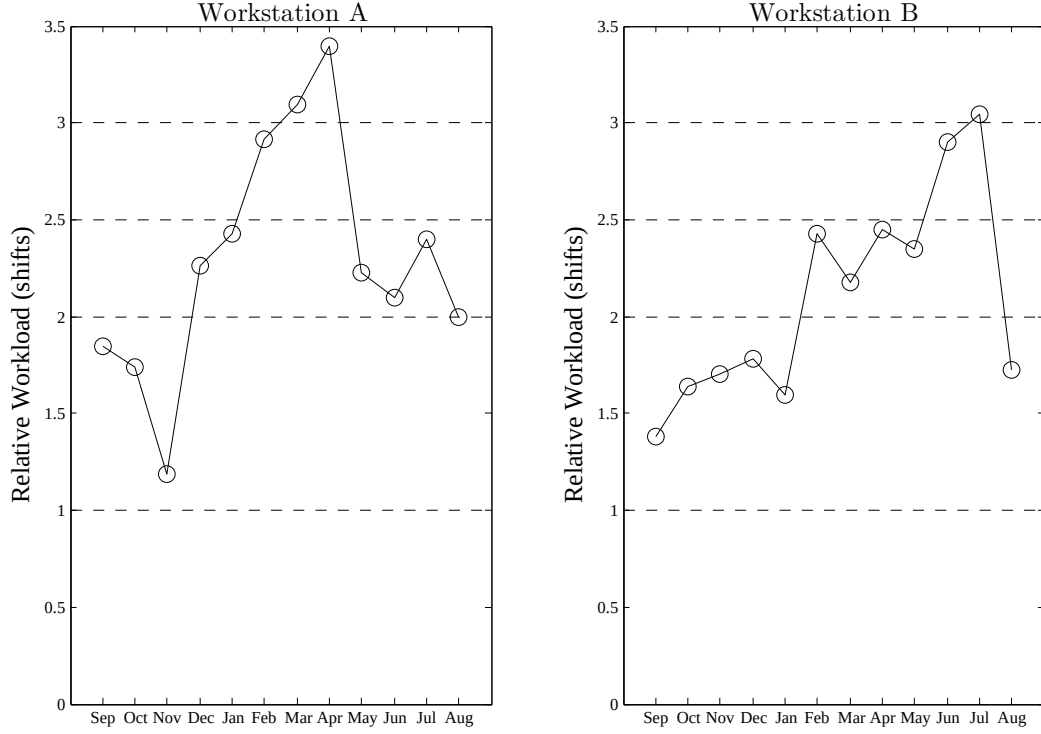


Figure 1: Monthly workload variation

A significant cause for the variations in workload is the underlying uncertainty in seasonal demands. Although, John Deere has a rough idea of the timing of their seasons, this knowledge does not necessarily mean that they know when each season would start, how long it would last, and how high demands would be in each season. These are often subject to uncertainties due to factors like weather, marketing strategies, state of economy etc. Further, the sequence of seasons is not always predictable. For instance, demands might switch between high, medium, and low demands in a random manner based on various factors. Sometimes, demands might skip a season altogether.

As demands are received from its customers, John Deere shares this demand and seasonality information with its manufacturing facilities and jointly develop strategies to market conditions. Often these strategies involve making investments in inventory stocking levels, or production capacities, or both. Adapting stocking levels to accommodate these demand changes can be costly since firms will need to stock significant volume of parts necessitating high investments in inventory holding costs, warehouse space, and risks of inventory obsolescence. Further, an aspect that often gets under-estimated is that efforts to ramp up to these stocking levels often leads to spikes in inven-

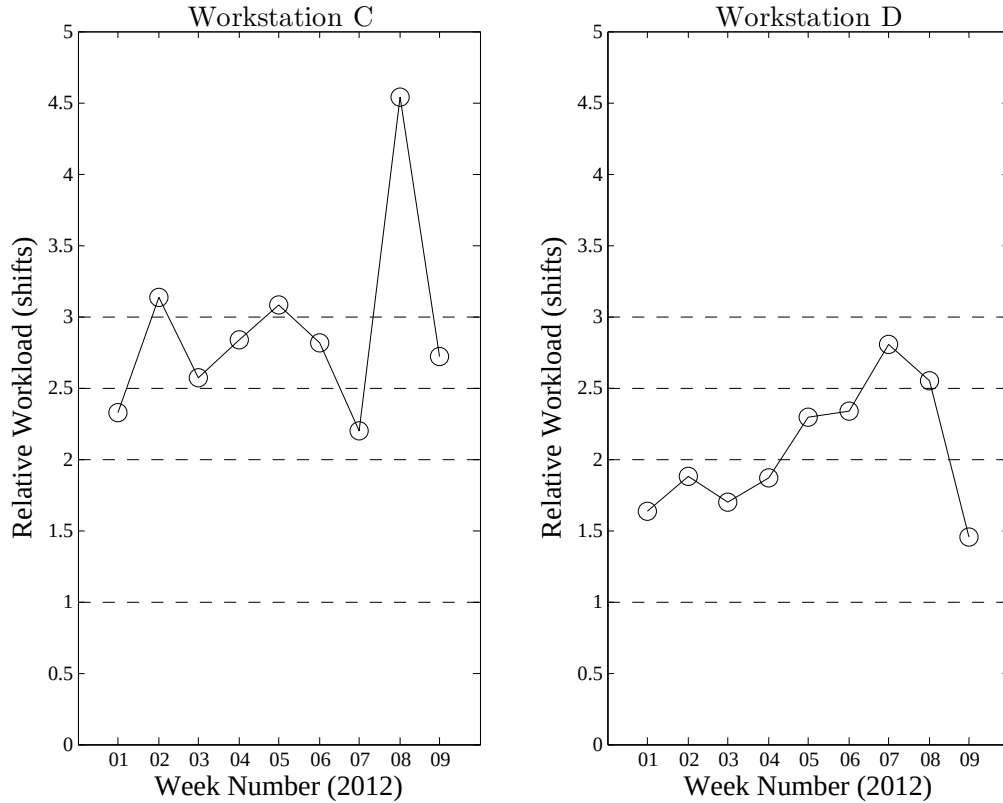


Figure 2: Weekly workload variation

tory replenishment lead times that in turn limit the manufacturers ability to reach the desired inventory level in a timely manner. As an alternative, some manufacturers pursue investments in capacity flexibility. However, increasing machine and labor capacity three-fold can be an expensive proposition (due to overtime costs and operator training costs). However, decreasing capacity also has cost implications in terms of workforce retention. Firms like John Deere, Harley-Davidson, and Electronic Theatre Controls, that go through these cycles on a regular basis have established flexible labor contracts that essentially allow them to operate their skilled workforce at various levels.

However, from an operational point of view, it is not clear how capacity and inventory flexibility must be leveraged to optimally respond to these various forms of seasonal demand trends. When do seasonal variations in demand have a significant impact? Are seasonal variations more critical when the variations are on a weekly time scale, monthly timescale or yearly timescale? What

attributes of seasonality are more critical to manufacturing operations: the variations in demands across seasons or the variations in lengths of the individual seasons, or both? When is it optimal to stock versus invest in capacity flexibility? Do optimal policies have a simple structure and how do optimal policies vary with various attributes of seasonality? How do the optimal policies compare with the policies used in industry? How inefficient are current practices?

We attempt to answer these questions in this research. Our contributions lie in providing the analytical foundations that provide structural insights into optimal policies for particular settings. In addition, we use our analytical framework to understand the various attributes of seasonality and the role of capacity and inventory flexibility in addressing these aspects of seasonal demands.

The rest of the paper is organized as follows. In the next section, Section 2 we review the relevant literature. In Section 3, we describe the production inventory system, and formulate the problem as a Markov decision process under finite horizon discounted expected cost, infinite horizon discounted expected cost, and infinite horizon average expected cost criteria. In Section 4, we identify the optimal solution and optimal policy under these settings. In Section 5, we obtain several properties of the optimal solution and derive an analytical expression of the optimal policy. In Section 6, we characterize the monotonic structure of the optimal policy. Section 7 conducts detailed study of the impact of various attributes of seasonal demands on the optimal policy and also compares the performance of the optimal policies with policies used in practice. These comparisons show where current policies can perform poorly. Section 8 discusses model extensions that relax some of the modeling assumptions. Section 9 summarizes the main conclusions.

2 Literature Review

There have been several studies that analyze the value of inventory flexibility in systems with seasonal demands. Song and Zipkin (1993) determine the optimal ordering policies to adopt stock levels to meet seasonal demand. They assume the replenishment lead time to be stochastic, but independent of order quantities. They show that when ordering cost are linear, the phase dependent base-stock policy is optimal, and when ordering cost are fixed, the phase dependent (s, S) policy is

optimal. Sethi and Cheng (1997) and Beyer and Sethi (1997) show that the (s, S) policy is optimal for inventory systems with fixed ordering cost under an external demand dependent on a Markov process. Kapuscinski and Tayur (1998) and Yossi and Federgruen (1997) analyze inventory ordering systems with bounded ordering size and constant lead times and show that a cyclic up-to level policy is optimal. Shang (2012) proposes a decomposition based heuristic for determining stocking levels in a serial inventory system under non-stationary demand. While these studies provide useful insights on optimal inventory policies in settings with seasonal demands, they do not study the impact of capacity flexibility under seasonal settings. Abhyankar and Graves (2001) consider an inventory system with MMPP demand process. They propose an intermediate-decoupling inventory policy, for determining inventory between stages of a manufacturing process, as a hedge for seasonal demands. Our work analyzes a production system that can respond to seasonal demands by adjusting *both* the production capacity *and* target inventory levels.

Angelus and Porteus (2002) analyze a manufacturing system to determine the optimal policy for capacity and production planning for short life cycle products under cyclic demands. They develop an optimal policy which provides a target interval of capacity levels for each time period. Hu et al. (2004) approximate the discrete state inventory variable as a continuous variable and obtain an optimal policy for manufacturing systems under seasonal demands. They assume that the arrival process and the production process are deterministic, and the arrival rate is required to be less than the production rate in each season. In practice, however, production rates and arrival rates have significant variability. Therefore, we model the production facility as a single server queue with exponentially distributed service times, and assume that the demand arrivals follow a Poisson process in each season with season dependent average demands. Moreover, we also allow the arrival rate of demands to exceed the production capacity in a season. These generalizations allow us to explore tradeoffs related to multiple aspects of seasonality. Gayon et al. (2009) study the production-inventory system with capacitated supply and MMPP demand. The system responds to the fluctuating demand through dynamic pricing and inventory replenishment. Kumar et al. (2013) analyze service facilities with exponentially distributed production times and seasonal demands. Using dynamic programming, they identify the optimal policy as a state dependent production rate.

The contributions of this research are as follows. First, for a manufacturing system that can invest in both capacity and inventory flexibility to deal with seasonal demands, we use analytical models to prove the existence of the optimal policy and solution. Subsequently, we prove the monotonicity of the optimal policy with respect to inventory levels, seasons, and workload levels. The monotonicity with workload levels emphasizes the impact of finite capacity and endogenous lead time on the optimal production inventory policies. As mentioned earlier, the impact of replenishment lead times on stocking levels needed to account for seasonal demands has significant practical importance. Third, we investigate the impact of different attributes of seasonality (such as the difference of average demand across seasons, variations in season length, and skewness in length and demand across seasons) on the optimal policy and the optimal cost. We also investigate the joint interaction between seasonality and system workload on the optimal policy and the optimal cost. Fourth, we compare our optimal policy with multiple simple and myopic policies that are sometimes adopted in practice. In particular, we compare the performance of the optimal policy to two other policies that make decisions only at the start of the seasons. These comparisons show the value of leveraging capacity and inventory flexibility dynamically. We also conduct comparisons with a myopic policy where the manufacturer makes decisions based only on the current season and ignores its impact on the subsequent seasons. These comparisons show the limitations of such a policy and illustrate the importance of considering the complete seasonal pattern in demand while developing the production and inventory control policies. Finally, we relax our modeling assumptions related to season duration and production cost function and analyze their impact on the structure of the optimal policy.

3 System Description and Model

We analyze a system that consists of a single manufacturing station and produced goods inventory buffer. The production time at the manufacturing station is exponentially distributed with mean μ^{-1} , implying a production rate of μ . We assume that the decision maker can vary μ between 0 and the maximum rate $\mu_{max} < \infty$. In practice, production rates are varied by adding extra working shifts, overtime production hours, or hiring temporary operators. The manufacturing system stocks

the produced goods in an inventory buffer. Arriving demands are satisfied immediately if there exists inventory in this buffer; else the demand waits as a backorder. We define the net inventory as the difference of the produced goods in the buffer and the backorders.

Customer demands are assumed to be non-stationary. In particular, the planning horizon is assumed to have m seasons and each season s , has a distinct average demand, λ_s . We model the seasonal demands as a Markov modulated Poisson process (MMPP) with m phases corresponding to the m seasons. This implies that demand arrivals in season $s \in \{1, \dots, m\}$ follow a Poisson process with arrival rate $\lambda_s < \infty$. A change in the season corresponds to a phase transition in the MMPP. This phase transition process is modeled as a finite state continuous time Markov chain with irreducible generator matrix Q that summarizes the transition probabilities between phases. The state space for this Markov chain is $\mathbb{S} := \{1, 2, \dots, m\}$. We assume that the phase transition process is ergodic and has unique stationary probability distribution vector denoted by $\{\hat{\pi}_1, \hat{\pi}_2, \dots, \hat{\pi}_m\}$. The decision maker observes the net inventory in the system and the arriving demands, and determines production rates and quantities to minimize total expected costs.

We choose to use MMPP framework to model seasonal demands since it allows us to capture various forms of uncertainty found in practice (for example, level of demand, duration and length of season, order of seasons) in a single framework. Further, the framework also provides a high degree of modeling flexibility - allowing us to analyze seasonal demand patterns that differ in the extent of uncertainty and variability (including processes that are close to being deterministic or have stochastic monotonicity) by suitably adjusting the governing matrix Q . In each of these settings, we investigate the optimal production inventory policy and the impact of seasonality on the policy and its performance. To do so, the dynamics of the system is modeled as a continuous-time Markov chain since the inter-arrival time, production time, and the duration of each demand season are exponentially distributed. We convert this into a discrete epoch Markov decision problem using uniformization technique, Lippman (1975). The Markov decision problem is formulated as follows:

Decision epochs: The decision epochs, $k = 1, 2, \dots$, correspond to instances of arrivals, production completion, and phase transition in the demand arrival process.

State space: The state space is modeled as a two dimensional Markov chain: $\mathbb{X} = \{(n, s) | n \in \mathbb{Z}, s \in \mathbb{S}\}$ where n is the net inventory of the system, and s is the demand phase. $\mathbb{Z}$ is the set of integers and $\mathbb{S} := \{1, 2, \dots, m\}$. Note that a positive value of n indicates availability of produced goods inventory and a negative value of n indicates backorders.

Action space: $\mathbb{A} = [0, \mu_{max}]$. The production rate can be varied between 0 and μ_{max} , and it is assumed that the maximum production rate is greater than the long run average arrival rate, i.e.

$$\mu_{max} > \sum_{s \in \mathbb{S}} \hat{\pi}_s \lambda_s \quad (1)$$

However, it is possible that λ_s could be greater than μ_{max} for some phase s . The condition in Equation 1 is used for ensuring the existence of the optimal cost under the infinite horizon average expected cost criterion.

Transition probabilities: The transition probability $p(n', s' | n, s, \mu)$ are expressed below:

$$\begin{aligned} p(n-1, s | n, s, \mu) &= \frac{\lambda_s}{\nu} & \forall n \in \mathbb{Z}, \forall s \in \mathbb{S} \\ p(n+1, s | n, s, \mu) &= \frac{\mu}{\nu} & \forall n \in \mathbb{Z}, \forall s \in \mathbb{S} \\ p(n, s' | n, s, \mu) &= \frac{Q_{ss'}}{\nu} & \forall n \in \mathbb{Z}, \forall s, s' \in \mathbb{S} \text{ and } s \neq s' \\ p(n, s | n, s, \mu) &= \left(1 - \frac{\lambda_s + \mu - Q_{ss}}{\nu}\right) & \forall n \in \mathbb{Z}, \forall s \in \mathbb{S} \end{aligned}$$

where $Q_{ss'}$ is the element of the generator matrix Q , $\nu = \lambda_{max} + \eta + \mu_{max}$, η is a positive scalar such that $\eta \geq \max\{-Q_{ss} | s \in \mathbb{S}\}$, and $\lambda_{max} = \max\{\lambda_1, \lambda_2, \dots, \lambda_m\}$.

Cost function: The cost function consists of production cost, inventory holding cost, and backorder cost. We assume linear inventory holding and backorder costs. The inventory holding cost per unit time per unit of inventory is h and the backorder cost per unit time per unit of backorder is b . The production cost is assumed to be piecewise linear function of the production rate μ . More specifically, the production cost increases linearly with rate c_{reg} up to μ_{reg} where $0 < \mu_{reg} < \mu_{max}$. Subsequently, the production cost increases linearly with rate c_{max} . In practice, c_{reg} could correspond to regular wage rate for employees as the production rate is varied between 0 and μ_{reg} . By incurring additional cost, the system can vary production rate by adjusting the number of employees. Any increase in the production rate beyond μ_{reg} requires the system to operate in the

secondary mode (second shift or overtime) that incurs a higher wage rate of c_{max} . The production cost per unit time $c(\mu)$, is represented as

$$c(\mu) = \begin{cases} c_{reg}\mu & 0 \leq \mu \leq \mu_{reg} \\ c_{reg}\mu_{reg} + c_{max}(\mu - \mu_{reg}) & \mu_{reg} < \mu \leq \mu_{max} \end{cases} \quad (2)$$

where $0 < c_{reg} < c_{max}$ are scalar constants. Note that the particular piece-wise linear cost structure described above is not a limiting feature of our model. We assume this structure solely for ease of illustration and our analysis can be extended to other general piece-wise linear and more general convex functions.

For the k^{th} transition, the cost function per unit time is expressed as $l_k(n, s, \mu) = c(\mu) + h(n)^+ + b(n)^-$, where $(x)^+ = \max(x, 0)$ and $(x)^- = \max(-x, 0)$. The discounted cost per transition is then obtained by integrating the discount factor $e^{-\alpha t}$ and $l_k(n, s, \mu)$ over the inter-transition time and integrating the inter-transition time over its probability distribution. The discounted cost function for the k^{th} transition, $l_{\alpha,k}(n, s, \mu)$, is given by:

$$\begin{aligned} l_{\alpha,k}(n, s, \mu) &= \int_0^\infty \left(\int_0^\tau l_k(n, s, \mu) e^{-\alpha t} dt \right) \nu e^{-\nu \tau} d\tau \\ &= \frac{1}{\alpha + \nu} l_k(n, s, \mu) \end{aligned}$$

In the next section, we formulate the objective function and show the existence of optimal solution and optimal policy.

4 Existence of Optimal Solution and Policy

We first formulate the problem under the discounted expected cost criterion, and then extend the formulation to the average expected cost criteria.

4.1 Discounted Expected Cost Criterion

Consider the finite horizon case. Let $v_{\alpha,k+1}(n, s)$ denote the finite horizon optimal discounted expected cost when $k + 1$ transitions are left. Then:

$$v_{\alpha,k+1}(n, s) = \inf_{\pi \in \Pi} \mathbb{E}_{(n,s)}^{\pi} \sum_{i=k+1}^1 l_{\alpha,i}(n', s', \mu_{\pi}) \quad (3)$$

The corresponding optimality equation is written as:

$$\begin{aligned} v_{\alpha,k+1}(n, s) = & \frac{1}{\alpha + \nu} \inf_{\mu \in \mathbb{A}} \left[l_{k+1}(n, s, \mu) + \mu v_{\alpha,k}(n+1, s) + \lambda_s v_{\alpha,k}(n-1, s) \right. \\ & \left. + \sum_{s' \in \mathbb{S}, s' \neq s} Q_{ss'} v_{\alpha,k}(n, s') + (\nu - \lambda_s - \mu + Q_{ss}) v_{\alpha,k}(n, s) \right] \\ & \text{for } n \in \mathbb{Z} \text{ and } s \in \mathbb{S} \end{aligned} \quad (4)$$

where $v_{\alpha,0}(n, s)$ is assumed 0 for all states. The first term in Equation 4 is the discounted cost incurred during the $(k+1)^{st}$ transition. The next three terms are the cost-to-go functions corresponding to production, arrival, and phase transition respectively. The last term in the equation is the uniformization term. Next, let $v_{\alpha}(n, s)$ denote the infinite horizon optimal discounted expected cost function. Then,

$$\begin{aligned} v_{\alpha}(n, s) &= \inf_{\pi \in \Pi} \lim_{k \rightarrow \infty} \mathbb{E}_{(n,s)}^{\pi} \sum_{i=k+1}^1 l_{\alpha,i}(n', s', \mu_{\pi}) \\ &= \inf_{\pi \in \Pi} \lim_{k \rightarrow \infty} v_{\alpha,k+1}^{\pi}(n, s) \end{aligned} \quad (5)$$

In order to show that Equation 5 holds, we need $\lim_{k \rightarrow \infty} v_{\alpha,k}(n, s) = v_{\alpha}(n, s)$. However, since our system has unbounded cost function per transition, we cannot directly apply the Banach fixed point theorem (Puterman, 1994) to show that $\lim_{k \rightarrow \infty} v_{\alpha,k}(n, s) = v_{\alpha}(n, s)$. Instead we show that $\lim_{k \rightarrow \infty} v_{\alpha,k}(n, s) = v_{\alpha}(n, s)$ in a modified space $\mathbb{V}_{\omega}$, using Theorem 11.5.3 of Puterman (1994). In Appendix A.1, we define this modified space, $\mathbb{V}_{\omega}$ and validate that the assumptions behind Theorem 11.5.3 hold for our system.

Applying Theorem 11.5.3 of Puterman (1994), then yields four key results for our system, namely i)

existence and uniqueness of the optimal solution ii) existence of a stationary optimal policy, iii) an optimal policy is any policy that minimizes the right side of Equation 6, and iv) $\lim_{k \rightarrow \infty} v_{\alpha,k}(n, s) = v_{\alpha}(n, s)$, listed below as Lemma 1.

Lemma 1. $\lim_{k \rightarrow \infty} v_{\alpha,k}(n, s) = v_{\alpha}(n, s)$ for $n \in \mathbb{Z}$ and for $s \in \mathbb{S}$.

In subsequent sections, we often prove results for the finite horizon case and then use Lemma 1 to extend these results to the infinite horizon case. For instance, applying the limit $k \rightarrow \infty$ to the finite horizon discounted expected cost optimal equation (Equation 4) and using Lemma 1, we obtain the following infinite horizon discounted cost optimal equation.

$$v_{\alpha}(n, s) = \frac{1}{\alpha + \nu} \min_{\mu \in \mathbb{A}} \left[l(n, s, \mu) + \mu v_{\alpha}(n+1, s) + \lambda_s v_{\alpha}(n-1, s) + \sum_{s' \in \mathbb{S}, s \neq s'} Q_{ss'} v_{\alpha}(n, s') \right. \\ \left. + (\nu - \lambda_s - \mu + Q_{ss}) v_{\alpha}(n, s) \right] \text{ for } n \in \mathbb{Z} \text{ and } s \in \mathbb{S} \quad (6)$$

Note that since we show the existence of stationary optimal policy (Appendix A.1), we are able to use min instead of inf in Equation 6.

4.2 Average Expected Cost Criterion

Next we present the analysis for the average expected cost criterion. The infinite horizon average expected cost of a policy π is

$$g^{\pi}(n, s) = \limsup_{K \rightarrow \infty} \frac{E_{\pi}(\sum_{k=0}^K l_k(n, s, \mu_{\pi}))}{K} \quad (7)$$

The optimal average expected cost is $g(n, s) = \inf_{\pi \in \Pi} g^{\pi}(n, s)$, and the policy π^* is optimal if $g(n, s) = g^{\pi^*}(n, s)$ for all $(n, s) \in \mathbb{Z} \times \mathbb{S}$. In fact, it can be shown that the optimal average expected cost is independent of the initial state, implying that $g(n, s)$ can be written as g (see Appendix A.2 for details). Let $w(n, s)$ be the corresponding relative value function. Then Equations 8 and 9 yield the values of g and $w(n, s)$ provided the limits exist.

$$g = \lim_{\alpha \downarrow 0} \alpha v_{\alpha}(n, s) \quad (8)$$

$$w(n, s) = \lim_{j \rightarrow \infty} (v_{\alpha_j}(n, s) - v_{\alpha_j}(n_0, s_0)) \quad (9)$$

where (n_0, s_0) is some distinguished state, and α_j is a converging sequence such that $\lim_{j \rightarrow \infty} \alpha_j = 0$. To obtain g and $w(n, s)$, we first subtract the discounted optimal cost $v_\alpha(n_0, s_0)$ from $v_\alpha(n, s)$ and subsequently take the limit $\alpha \rightarrow 0$. We get:

$$\begin{aligned}
v_\alpha(n, s) - v_\alpha(n_0, s_0) &= \frac{1}{\alpha + \nu} \min_{\mu \in \mathbb{A}} \left[l(n, s, \mu) - \alpha v_\alpha(n_0, s_0) + \mu[v_\alpha(n+1, s) - v_\alpha(n_0, s_0)] \right. \\
&\quad + \lambda_s[v_\alpha(n-1, s) - v_\alpha(n_0, s_0)] + \sum_{s' \in \mathbb{S}, s \neq s'} Q_{ss'}[v_\alpha(n, s') - v_\alpha(n_0, s_0)] \\
&\quad \left. + (\nu - \lambda_s - \mu + Q_{ss})[v_\alpha(n, s) - v_\alpha(n_0, s_0)] \right] \quad \text{for } n \in \mathbb{Z} \text{ and } s \in \mathbb{S} \quad (10)
\end{aligned}$$

We show that the limits in Equations 8 and 9 exist by applying Theorem 1 of Sennott (1989). In Appendix A.2 we show that the system satisfies the necessary assumptions (Assumptions, 4, 5, 6, and 7) needed to apply this theorem and also explain the implications of these assumptions in the context of our system. Applying Theorem 1 of Sennott (1989) we get the following inequality:

$$\begin{aligned}
w(n, s) + g &\geq \frac{1}{\nu} \min_{\mu \in \mathbb{A}} [l(n, s, \mu) + \mu w(n+1, s) + \lambda_s w(n-1, s) + \sum_{s' \in \mathbb{S}} Q_{ss'} w(n, s') \\
&\quad + (\nu - \lambda_s - \mu) w(n, s)] \quad \text{for } n \in \mathbb{Z} \text{ and for } s \in \mathbb{S} \quad (11)
\end{aligned}$$

The theorem also guarantees the existence of a stationary average expected cost optimal policy and enables us to extend any property of the discounted cost optimal policy to the average cost optimal policy.

We thus establish the existence of a unique optimal solution and a stationary optimal policy under both discounted expected cost and average expected cost criteria. In the next section, we derive properties of the optimal discounted expected cost and determine an analytical expression for the discounted expected cost optimal policy.

5 Properties of the Optimal Solution and Optimal Policy

This section derives properties of the optimal solution and optimal policy. In Section 5.1, we show that the optimal cost function, $v_\alpha(n, s)$ is a convex function of n for a fixed s . Note that, convexity of the optimal cost function implies that the first difference $y_\alpha(n, s) = v_\alpha(n, s) - v_\alpha(n-1, s)$ is non-decreasing with respect to n for a fixed s . We then prove the asymptotic properties of $y_\alpha(n, s)$, and subsequently derive an analytical expression for the optimal policy in Section 5.2. Using asymptotic properties of $y_\alpha(n, s)$ and the analytical expression of the optimal policy, we identify special cases under which the flexibility in the production rate is more valuable. In later sections, we use the monotonic property of $y_\alpha(n, s)$ to show the structural properties of the optimal policy.

5.1 Properties of the Optimal Cost Function

Proposition 1. *For each $s \in \mathbb{S}$, $v_\alpha(n, s)$ is a convex function of n .*

We prove the convexity result in Proposition 1 by showing that $y_\alpha(n, s)$ is non-decreasing in n for each $s \in \mathbb{S}$. Let $y_{\alpha,k+1}(n, s) = v_{\alpha,k+1}(n, s) - v_{\alpha,k+1}(n-1, s)$. Then, from Equation 6:

$$\begin{aligned} y_{\alpha,k+1}(n, s) &= \frac{1}{\alpha + \nu} \left[h1_{\{n>0\}} - b1_{\{n<0\}} + \lambda_s y_{\alpha,k}(n-1, s) + \sum_{s' \in \mathbb{S}, s \neq s'} Q_{ss'} y_{\alpha,k}(n, s') \right. \\ &\quad \left. + (\nu - \lambda_s + Q_{ss}) y_{\alpha,k}(n, s) + \min_{\mu \in \mathbb{A}} [c(\mu) + \mu y_{\alpha,k}(n+1, s)] \right. \\ &\quad \left. - \min_{\mu \in \mathbb{A}} [c(\mu) + \mu y_{\alpha,k}(n, s)] \right] \quad \text{for } n \in \mathbb{Z} \text{ and for } s \in \mathbb{S} \end{aligned} \quad (12)$$

where $1_{\{x\}}$ equals 1 if x is true and 0 otherwise. Using induction, we first show $y_{\alpha,k}(n, s)$ is non-decreasing in n and then extend the result to $y_\alpha(n, s)$ using Lemma 1. The proof of Proposition 1 is in Appendix A.3. Next, using the monotonicity of $y_\alpha(n, s)$ in n , we prove the following asymptotic result.

Proposition 2. *For a fixed $s \in \mathbb{S}$,*

1. $y_\alpha(n, s) \rightarrow \frac{-b}{\alpha}$ as $n \rightarrow -\infty$
2. $y_\alpha(n, s) \rightarrow \frac{h}{\alpha}$ as $n \rightarrow +\infty$

Since $y_\alpha(n, s)$ is a non-decreasing function of n , to prove Proposition 2, we show $\frac{-b}{\alpha}$ is the greatest lower bound and $\frac{b}{\alpha}$ is the least upper bound of $y_\alpha(n, s)$. The details of the proof are in Appendix A.4. Proposition 2 leads to Proposition 3 that yields another important property of $v_\alpha(n, s)$.

Proposition 3. *For each $s \in \mathbb{S}$, there exist a finite $n_{o,s} \in \mathbb{Z}$ such that $v_\alpha(n, s)$ is decreasing in n at the fixed s for all $-\infty < n < n_{o,s}$, and increasing in n for all $n_{o,s} \leq n < \infty$.*

We prove the result by contradiction and the details are provided in Appendix A.5.

5.2 Analytical Expression for an Optimal Policy

Recall that the policy which minimizes the right side of Equation 6 is an optimal policy. By minimizing the right side of Equation 6, we obtain a structured optimal policy which we state as Theorem 1.

Theorem 1. *The infinite horizon discounted cost optimal equation (Equation 6) has a structured optimal policy characterized by finite inventory levels $\{z_{\mu,s}, z_{b,s}\}$ for each season $s \in \mathbb{S}$. The optimal production rate $\mu^*(n, s) = \mu_{max}$ for $n < z_{\mu,s}$, $\mu^*(n, s) = \mu_{reg}$ for $z_{\mu,s} \leq n < z_{b,s}$, and $\mu^*(n, s) = 0$ for $z_{b,s} \leq n$.*

Proof. We choose the μ dependent terms from right side of Equation 6 and apply the first order minimization condition to obtain the expression for the optimal policy $\mu^*(n, s)$, i.e.

$$\mu^*(n, s) = \arg \min_{\mu \in \mathbb{A}} \left(c(\mu) + \mu y_\alpha(n+1, s) \right) \quad (13)$$

From Equation 13 and by exploiting the piece-wise linearity of the production cost function (Equation 2), we obtain the optimal policy as a function of $y_\alpha(n+1, s)$ as shown in Table 1.

Table 1: Optimal policy as a function of $y_\alpha(n+1, s)$

$y_\alpha(n+1, s)$	$\mu^*(n, s)$
$-c_{reg} < y_\alpha(n+1, s)$	0
$-c_{max} < y_\alpha(n+1, s) \leq -c_{reg}$	μ_{reg}
$y_\alpha(n+1, s) \leq -c_{max}$	μ_{max}

By taking inverse of $y_\alpha(., s)$, we obtain the optimal policy for the entire range of the net inventory n for a given s . For ease of notation, we replace $y_\alpha(n+1, s)$ with y in the rest of this subsection.

Since the function y is not strictly monotone, its inverse may have multiple entries. To meet the conditions in the first column of Table 1, we choose the maximum of these entries. Denote $\lfloor \max y^{-1}(-c_{max}) \rfloor$ as $z_{\mu,s}$ and $\lfloor \max y^{-1}(-c_{reg}) \rfloor$ as $z_{b,s}$. This allows us to express the optimal policy using $z_{\mu,s}$, $z_{b,s}$, in Table 2. $\square$

Table 2: Optimal policy as function of n

n	$\mu^*(n, s)$
$z_{b,s} \leq n$	0
$z_{\mu,s} \leq n < z_{b,s}$	μ_{reg}
$n < z_{\mu,s}$	μ_{max}

The optimal policy suggests that in a given phase, the facility should continue producing till the net inventory hits the level $z_{b,s}$. The optimal policy also suggests to set the production rate to μ_{reg} when the net inventory is between $[z_{\mu,s}, z_{b,s})$ and to μ_{max} when the net inventory is less than $z_{\mu,s}$. Note that, the piece-wise linear structure of the production cost function expressed in Equation 2 leads to only two optimal production rates μ_{reg} and μ_{max} . For illustration, in Figure 3 we plot $y_\alpha(n, s)$ as function of n for a system with three phases at $s = 3$. For this setting, the optimal threshold levels for phase 3 are $[z_{\mu,3}, z_{b,3}] = [18, 35]$ for phase 3, implying that the system process at rate $\mu_{max} = 15$ for net inventory level below 18, at rate $\mu_{reg} = 10$ for net inventory level up to 34, and stop production at net inventory level 35.

The optimal policy in Theorem 1 is a phase dependent base-stock policy with state dependent production rate. In phase s , the base-stock level equals $z_{b,s}$ and the net inventory level at which the system should switch the production rate from μ_{reg} to μ_{max} or vice versa is $z_{\mu,s}$. Next, in the following Corollary, we show that the threshold levels are always finite whenever $c_{max} < \frac{b}{\alpha}$, and also discuss the finiteness of the threshold levels at other relative values of c_{max} , c_{reg} , and b .

Corollary 1. *The base-stock levels, $z_{\mu,s}$ and $z_{b,s}$ satisfy the following properties for each $s \in \mathbb{S}$:*

1. *If $c_{max} < \frac{b}{\alpha}$ then $-\infty < z_{\mu,s}, z_{b,s} < \infty$.*
2. *If $c_{reg} < \frac{b}{\alpha} \leq c_{max}$, then $z_{\mu,s} = -\infty$, and $-\infty < z_{b,s} < \infty$.*
3. *If $c_{reg} \geq \frac{b}{\alpha}$, then $z_{\mu,s} = -\infty$, and $z_{b,s} = -\infty$.*

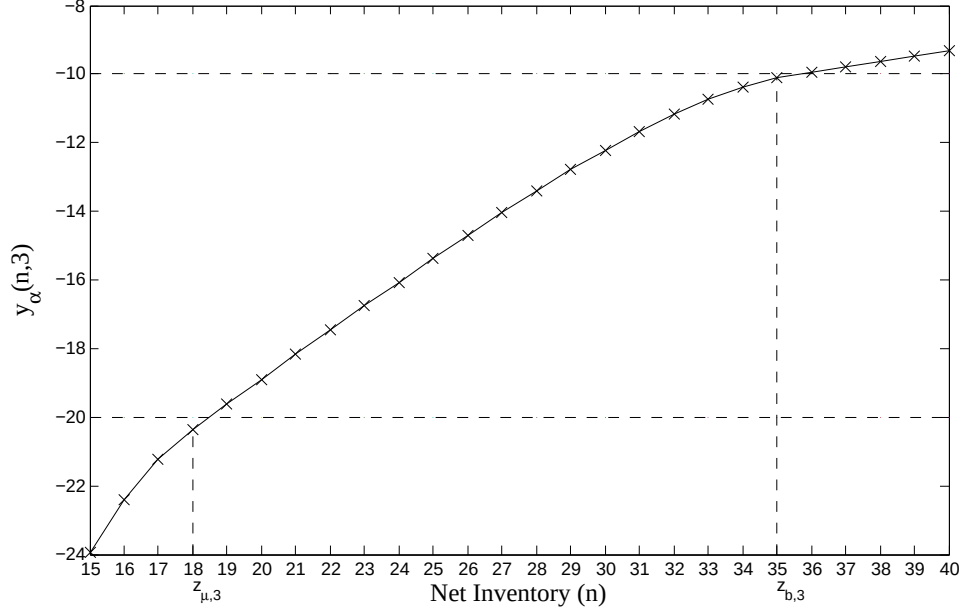


Figure 3: An illustrative example

Proof. When $c_{max} < \frac{b}{\alpha}$, then $\frac{-b}{\alpha} < -c_{max} < \frac{h}{\alpha}$, and $\frac{-b}{\alpha} < -c_{reg} < \frac{h}{\alpha}$. Using arguments similar to the proof of Proposition 3, we show that the optimal inventory threshold levels $z_{\mu,s}$ and $z_{b,s}$ are finite, and also bounded above by $n_{o,s}$ since $-c_{max} < 0$ and $-c_{reg} < 0$. Parts 2 and 3 of Corollary 1 are proved similarly. $\square$

Thus, when $c_{max} < \frac{b}{\alpha}$, the manufacturer can avail the flexibility in the production rate, by switching to either μ_{reg} and μ_{max} at appropriate inventory levels. When $c_{reg} < \frac{b}{\alpha} \leq c_{max}$, production is undertaken only at the rate μ_{reg} , and never at μ_{max} . In this case, the manufacturer has to run the production at a fixed rate μ_{reg} and flexibility in the production rate is of no value. Finally, $c_{reg} \geq \frac{b}{\alpha}$, is a trivial case where it is never optimal to undertake the production at any production rate.

6 Monotonicity of the Optimal Policy

In this section, we establish the monotonicity of the optimal policy with respect to the net inventory, n , the demand phase, s , and the workload level.

6.1 Monotonicity with respect to Net Inventory Level

The following theorem establishes the monotonicity of the optimal policy under both discounted and average cost criteria.

Theorem 2. *For the system described in Section 3:*

1. *The optimal policy $\mu_\alpha^*(n, s)$ is non-increasing in $y_\alpha(n+1, s)$ for each $(n, s) \in \mathbb{Z} \times \mathbb{S}$*
2. *There exists a discounted expected cost optimal policy $\mu_\alpha^*(n, s)$ which is non-increasing in n for each $s \in \mathbb{S}$*
3. *There exists an average expected cost optimal policy $\mu^*(n, s)$ which is non-increasing in n for each $s \in \mathbb{S}$*

Proof. From Table 1, it is evident that $\mu_\alpha^*(n, s)$ is non-increasing in $y_\alpha(n+1, s)$. Further, from Proposition 1, we know that $y_\alpha(n+1, s)$ is non-decreasing with respect to n . Hence, $\mu_\alpha^*(n, s)$ is non-increasing in n , which proves Part 2 of the Proposition 2. To prove Part 3, we choose a subsequence of discount factors $\alpha_i \rightarrow 0$ such that the discounted expected cost optimal policy $\mu_{\alpha_i}^*(n, s)$ converge to average expected cost optimal policy $\mu^*(n, s)$ (see Theorem 1 of Sennott (1989) and Appendix A.2). Thus, $\lim_{\alpha_i \rightarrow 0} \mu_{\alpha_i}^*(n, s) \geq \lim_{\alpha_i \rightarrow 0} \mu_{\alpha_i}^*(n+1, s)$ leads to $\mu^*(n, s) \geq \mu^*(n+1, s)$ for $n \in \mathbb{Z}$ and $s \in \mathbb{S}$. $\square$

6.2 Monotonicity with respect to Demand Phase

Theorem 3 shows that the optimal policy is monotone with respect to the demand phase, s when the average demand λ_s increases with phase s and the phase transition process is stochastically monotone (as defined in Keilson and Kester (1977)).

Theorem 3. *If the phase transition process is stochastically monotone and $\lambda_1 < \lambda_2 < \dots < \lambda_m$, then:*

1. *There exists a discounted expected cost optimal policy $\mu_\alpha^*(n, s)$ which is non-decreasing in s for each $n \in \mathbb{Z}$*

2. *There exists an average expected cost optimal policy $\mu^*(n, s)$ which is non-decreasing in s for each $n \in \mathbb{Z}$*

Proof outline: From part 1 of Theorem 2, $\mu_\alpha^*(n, s)$ is non-increasing in $y_\alpha(n+1, s)$. We show part 1 of Theorem 3 by proving that $y_\alpha(n+1, s)$ is non-increasing with respect to s for each $n \in \mathbb{Z}$. Part 2 of Theorem 3 follows by applying Theorem 1 of Sennott (1989). Details of the proof are in Appendix A.6.

6.3 Monotonicity with respect to Workload

In our analysis, we explicitly model congestion delays and its impact on production lead time at various workloads. Thus, the optimal threshold levels $\{z_{\mu, s}, z_{b, s}\}$ vary with the workload level. In this section, we show the structural relation between the optimal policy and workload levels. We use the ratio of phase averaged demand and maximum production rate. i.e., $\frac{\sum_{s \in \mathbb{S}} \hat{\pi}_s \lambda_s}{\mu_{max}}$ as a measure of the workload in the system.

Theorem 4. *For $\epsilon \geq 0$, if $\mu_{\alpha, \epsilon}^*(n, s)$ is a discounted cost optimal policy for a system with demand $\Lambda_\epsilon = [\lambda_1 + \epsilon, \dots, \lambda_m + \epsilon]$ and $\mu_{max} > \sum_{s \in \mathbb{S}} \hat{\pi}_{s, \epsilon}(\lambda_s + \epsilon)$, then $\mu_{\alpha, \epsilon}^*(n, s)$ is non-decreasing with ϵ .*

Proof outline: To prove Theorem 4, we use Proposition 2 to show that $\mu_\alpha^*(n, s) \leq \mu_{\alpha, \epsilon}^*(n, s)$ for all $n \in \mathbb{Z}$ and $s \in \mathbb{S}$ where $\mu_\alpha^*(n, s)$ is the optimal policy for a system with $\Lambda = [\lambda_1, \dots, \lambda_m]$. Details of the proof are in Appendix A.7.

Theorem 4 implies that as the workload on the system increases, the optimal policy is non-decreasing at all phases and net inventory levels.

7 Numerical Study

In this section, we investigate the impact of different seasonal conditions and workload levels on the optimal cost and optimal policy. Section 7.1 investigates the impact of season length and demand variation across seasons on the optimal policy. Section 7.2 investigates the impact of skewed seasonality, where demand in the peak season might exceed maximum production capacity. Section 7.3 investigates the impact of workload on optimal cost and optimal policy. Finally, in Section

7.4, we examine how the performance of the optimal policy compares with other commonly used policies. We first describe the parameter settings common to these experiments before we discuss the particular findings.

We consider a manufacturing system where μ_{reg} is set to 10 units per day and μ_{max} is set to 15 units per day. We assume that the demand has three seasons modeled by a three-phase MMPP. The generator matrix Q for the MMPP is assumed as

$$Q = \begin{bmatrix} -q & q & 0 \\ q & -2q & q \\ 0 & q & -q \end{bmatrix}$$

where the average duration of each season is $\frac{1}{q}$. Note that, the structure of matrix Q is such that, the phase transition process is a stochastically monotone process. The average demand λ_1, λ_2 , and λ_3 in seasons 1, 2 and 3 are 9, 11 and 13 units per day respectively. This implies that the season-averaged demand $\bar{\lambda} = \sum_{s \in \mathbb{S}} \hat{\pi}_s \lambda_s$ is 11 units per day. The holding cost h equals \$2 per unit per day and the backorder cost b equals \$50 per unit per day. The production costs are $c_{reg} = \$10$ and $c_{max} = \$20$. These parameters are summarized in Table 3. In our numerical experiments, we consider several variations of this base setting.

Table 3: System parameters and cost parameters

System Parameters		Cost Parameters	
μ_{reg}, μ_{max}	10, 15	c_{reg}, c_{max}	10, 20
$[\lambda_1, \lambda_2, \lambda_3]$	[9,11,13]	h, b	2, 50

The average cost and the threshold levels of the policies are estimated using policy iteration algorithm under the infinite horizon average cost criterion. For computational purposes, we truncate the state space to [-600:600] i.e. from 600 units of backorder through 600 units of on-hand inventory.

7.1 Impact of Season Lengths and Demand Variation

Demand is considered to be more seasonal when, either the difference in the average demand across the seasons increases or when the average season length increases (i.e. q decreases). To study the impact of these attributes of seasonality, we evaluate the optimal threshold levels under nine cases summarized in Table 4. The nine cases correspond to three values of q (0.1, 0.05, 0.01) and three settings of average demand ([9,11,13], [8,11,14], [7,11,15]). The values of q correspond to average season duration of 10 days, 20 days, and 100 days. In the demand settings, we keep the average demand in season 2 to 11 units per day, and vary the demand in other seasons (season 1 and season 3) by 2 units per day. The optimal policy for the nine cases are listed in the last column of Table 4. For each case, the optimal policy is presented in the form of a 2×3 matrix where the first row corresponds to $[z_{\mu,1} \ z_{\mu,2} \ z_{\mu,3}]$ and the second row corresponds to $[z_{b,1} \ z_{b,2} \ z_{b,3}]$.

Table 4: Impact of season length and demand variations

Cases	Average Season Length $1/q$	Demand $[\lambda_1, \lambda_2, \lambda_3]$	Optimal Average Cost g	Δ_s	Optimal Policy $\begin{bmatrix} z_{\mu,1} & z_{\mu,2} & z_{\mu,3} \\ z_{b,1} & z_{b,2} & z_{b,3} \end{bmatrix}$
1	10	[9,11,13]	3.303	9.99	$\begin{bmatrix} 3 & 8 & 15 \\ 15 & 25 & 36 \end{bmatrix}$
2		[8,11,14]	3.821	27.24	$\begin{bmatrix} 2 & 12 & 24 \\ 14 & 32 & 49 \end{bmatrix}$
3		[7,11,15]	4.912	63.57	$\begin{bmatrix} 3 & 25 & 41 \\ 22 & 47 & 71 \end{bmatrix}$
4	20	[9,11,13]	3.306	10.09	$\begin{bmatrix} 2 & 8 & 17 \\ 13 & 24 & 39 \end{bmatrix}$
5		[8,11,14]	3.896	29.74	$\begin{bmatrix} 1 & 13 & 29 \\ 11 & 32 & 57 \end{bmatrix}$
6		[7,11,15]	5.541	84.52	$\begin{bmatrix} 2 & 35 & 58 \\ 19 & 57 & 91 \end{bmatrix}$
7	100	[9,11,13]	3.289	9.52	$\begin{bmatrix} 2 & 7 & 19 \\ 11 & 22 & 44 \end{bmatrix}$
8		[8,11,14]	3.943	31.30	$\begin{bmatrix} 1 & 10 & 39 \\ 9 & 29 & 70 \end{bmatrix}$
9		[7,11,15]	7.970	165.40	$\begin{bmatrix} 0 & 76 & 130 \\ 10 & 103 & 167 \end{bmatrix}$

To understand the impact of seasonality, we compare the nine cases in Table 4 with the sta-

tionary demand case [11 11 11]. For the stationary demand case, the optimal average cost is 3.003, and the optimal threshold levels are $\begin{bmatrix} 6 & 6 & 6 \\ 22 & 22 & 22 \end{bmatrix}$. Denote the percentage difference between the average cost of the seasonal demand and that of the stationary demand as Δ_s i.e. $\Delta_s = \frac{Cost_{seasonal} - Cost_{stationary}}{Cost_{stationary}} \times 100$, where $Cost_{stationary} = 3.003$. For each of the nine cases, the optimal cost, Δ_s , and the optimal policy are listed in Table 4.

For a fixed q , as the difference of the average demand among the seasons increases, the optimal average cost increases, and Δ_s increases to as high as 165%. Thus, the seasonality in the demand can increase the production and inventory cost significantly when compared to the stationary demand case even when the phase-averaged workload levels are the same. Further, for a fixed demand profile, the optimal average cost increases as season length increases (q decreases). Thus, seasonality in the demand (high difference of average demand among seasons and/or long seasons) increases the average cost, because the manufacturer either has to build inventory in advance for peak demand season which increases the holding cost or has to ramp up the production capacity during peak demand seasons which increases the production cost.

We next focus on the impact of seasonality on the optimal threshold levels. Recall that for the stationary case, the optimal threshold levels are $\begin{bmatrix} 6 & 6 & 6 \\ 22 & 22 & 22 \end{bmatrix}$. For a fixed q , as the difference of the average demand among the seasons increases, the threshold levels of peak season (season 3) increase and that of low season (season 1) decrease compared to the respective thresholds of the stationary demand case. The threshold levels of season 2, which has the same average demand as season 2 of stationary demand case, increase with increase in the difference of the average demand among seasons. The threshold levels in season 1 (3) decrease (increase) primarily due to decrease (increase) in the average demand in that season. Threshold levels of season 2 increase in anticipation of peak demand in season 3. For a given demand profile, the increase in threshold levels of seasons 2 and 3, or decrease in threshold levels of season 1 is significant at higher season lengths. Further, the behavior of the threshold levels $z_{\mu,i}$ illustrate the value of capacity flexibility. For instance, when $1/q = 100$, we notice that $z_{\mu,3}$ increases from 19 to 130 as the demand variability

increases. To respond to increased demand in season 3, the system leverages the capacity flexibility to operate the system at a higher capacity and build more inventory. This effect is also observed for other values of $1/q$.

7.2 Impact of Skewed Seasonality

In many practical settings, demands exhibit skewed seasonal demand patterns, i.e. the demand in the peak season exceeds the maximum production rate, but the duration of the peak season is shorter compared to other seasons. To understand the structure of the optimal policy in such settings, we consider a scenario where the lengths of the individual seasons are in the ratio of 3:2:1, i.e. season 1 with low average demand has the longest duration, and season 3 with peak demand has the shortest duration (i.e. $\frac{1}{q_1} > \frac{1}{q_2} > \frac{1}{q_3}$). We consider three cases where the planning horizon ($\sum_S \frac{1}{q_s}$) is 30 days, 60 days, and 300 days, respectively. We set the demand in each season so that the season-averaged workload is 11 units per day. Further, we set the average demand in season 2 to be equal to 11 units per day, the average demand in season 1 to be equal to 9 units per day, i.e. 2 units below season 2 demand, and the average demand in season 3 to be equal to 17 units per day, yielding a season averaged demand of 11 units per day. Note that demand in season 3 is 2 units above the maximum production capacity, $\mu_{max} = 15$ units per day. We compare the three cases listed in Table 5 with the stationary demand case, [11 11 11]. The optimal average cost for the stationary demand case is 3.003 and the optimal policy is $\begin{bmatrix} 6 & 6 & 6 \\ 22 & 22 & 22 \end{bmatrix}$.

We observe that when compared to the stationary demand case, skewness in the seasonality increases the production and inventory management cost and this impact is significantly higher at longer planning horizon levels. Further, compared to the stationary case, the threshold levels are higher in all seasons under the skewed seasonal demand conditions, and the increase in the threshold levels are significant at higher planning horizon levels. We observe that even though season 3 has a relative short duration compared to the other seasons, due to high demand, the optimal policy recommends building up inventories in seasons 1 and 2, resulting in higher threshold levels in these seasons.

Table 5: Impact of skewed seasonality

Cases	Horizon $\sum_{\mathbb{S}} \frac{1}{q_s}$	Transition matrix Q	Demand $[\lambda_1, \lambda_2, \lambda_3]$	Optimal Cost g	Δ_s	Optimal Policy $\begin{bmatrix} z_{\mu,1} & z_{\mu,2} & z_{\mu,3} \\ z_{b,1} & z_{b,2} & z_{b,3} \end{bmatrix}$
1	30	$\begin{bmatrix} -1/15 & 1/15 & 0 \\ 1/10 & -2/10 & 1/10 \\ 0 & 1/5 & -1/5 \end{bmatrix}$	[9, 11, 17]	5.351	78.19	$\begin{bmatrix} 9 & 39 & 58 \\ 31 & 63 & 89 \end{bmatrix}$
2	60	$\begin{bmatrix} -1/30 & 1/30 & 0 \\ 1/20 & -2/20 & 1/20 \\ 0 & 1/10 & -1/10 \end{bmatrix}$	[9, 11, 17]	7.092	136.16	$\begin{bmatrix} 14 & 70 & 103 \\ 40 & 96 & 139 \end{bmatrix}$
3	300	$\begin{bmatrix} -1/150 & 1/150 & 0 \\ 1/100 & -1/50 & 1/100 \\ 0 & 1/50 & -1/50 \end{bmatrix}$	[9, 11, 17]	20.198	572.59	$\begin{bmatrix} 68 & 309 & 439 \\ 112 & 352 & 483 \end{bmatrix}$

7.3 Impact of Workload Levels

Next, we investigate whether seasonality effects are more dominant at higher system workloads. We use $\frac{\bar{\lambda}}{\mu_{max}}$ as a measure of workload on the manufacturing system, and evaluate the optimal average cost and policy as the workload varies from 70% to 90%. We set $q = 0.05$, $\mu_{reg} = 10$, $\mu_{max} = 15$, $\lambda_1 = \bar{\lambda} - 4$, $\lambda_2 = \bar{\lambda}$, and $\lambda_3 = \bar{\lambda} + 4$ units. The cost parameters are the same as in Table 3. Table 6 summarizes the results. In Table 6, we observe that as the workload level increases,

Table 6: Impact of workload levels

Cases	Work-load	Demand $[\lambda_1, \lambda_2, \lambda_3]$	Optimal Cost g	Δ_s	Optimal Policy $\begin{bmatrix} z_{\mu,1} & z_{\mu,2} & z_{\mu,3} \\ z_{b,1} & z_{b,2} & z_{b,3} \end{bmatrix}$	Optimal Policy Stationary Case
1	70%	[6.5, 10.5, 14.5]	4.332	59.38	$\begin{bmatrix} 0 & 19 & 40 \\ 9 & 38 & 70 \end{bmatrix}$	$\begin{bmatrix} 5 & 5 & 5 \\ 18 & 18 & 18 \end{bmatrix}$
2	75%	[7.25, 11.25, 15.25]	6.371	101.55	$\begin{bmatrix} 6 & 47 & 71 \\ 29 & 70 & 105 \end{bmatrix}$	$\begin{bmatrix} 7 & 7 & 7 \\ 24 & 24 & 24 \end{bmatrix}$
3	80%	[8, 12, 16]	10.102	171.49	$\begin{bmatrix} 51 & 101 & 131 \\ 79 & 130 & 169 \end{bmatrix}$	$\begin{bmatrix} 11 & 11 & 11 \\ 31 & 31 & 31 \end{bmatrix}$
4	85%	[8.75, 12.75, 16.75]	16.609	272.98	$\begin{bmatrix} 144 & 200 & 236 \\ 174 & 236 & 278 \end{bmatrix}$	$\begin{bmatrix} 17 & 17 & 17 \\ 41 & 41 & 41 \end{bmatrix}$
5	90%	[9.5, 13.5, 17.5]	28.516	414.92	$\begin{bmatrix} 327 & 387 & 431 \\ 360 & 431 & 477 \end{bmatrix}$	$\begin{bmatrix} 28 & 28 & 28 \\ 57 & 57 & 57 \end{bmatrix}$

Δ_s increases significantly. This increase is primarily due to the increase replenishment lead times that increases the expected backorder costs. Further, at higher workload levels, the threshold levels increase significantly compared to the stationary demand case. The threshold levels are higher because the system builds up inventory in seasons with low demand to satisfy demands in the

seasons with high demand.

7.4 Comparison of Optimal Policy with Other Policies

In this section, we compare the average cost of the optimal policy with other suboptimal policies that are used in practice. We study three types of policies.

- Policy 1: Production rate equals μ_{reg} in season 1 and μ_{max} in seasons 2 and 3.
- Policy 2: Production rate equals μ_{max} in all the seasons.
- Policy 3: Myopic policy where threshold levels for each season are determined ignoring the effect of other seasons.

In the first policy, the base-stock level varies with seasons, but the production rate is set to μ_{reg} in season one and μ_{max} in seasons two and three. In the second policy, the base-stock level depends on the season, but the production rate is set to μ_{max} in all the seasons. The optimal base-stock levels for both policies are identified using the policy iteration algorithm under the infinite horizon average cost criterion. The practical advantages of these policies are that the change in the production rate only occurs when the season changes. The third policy is a myopic policy where the threshold levels in each season are estimated by assuming the demand as stationary (non-seasonal) with value equal to the average demand of the season. For instance, if the demand vector is [9, 11, 13] then threshold levels for season 1, 2, and 3 are obtained by assuming that the demand is a stationary Poisson process with parameters of 9, 11, and 13 units/day respectively. If the demand in any season exceeds the maximum production rate, then the myopic policy builds up to the maximum possible inventory level at the production rate μ_{max} . Once the threshold levels are obtained for each season, then the average cost for the myopic policy is evaluated under the infinite horizon average cost criterion.

Let Δ_i denote the percentage of average cost difference between these policies and the optimal policy. The percentage cost difference for policy $i \in \{1, 2, 3\}$, Δ_i equals $\frac{Cost_i - Cost_o}{Cost_o} \times 100$ where $Cost_i$ is the average cost for policy i and $Cost_o$ is the average cost under the optimal policy. We compare the three policies with the optimal policy for the cases listed in Tables 4 and 5. The results

are listed in Tables 7, 8, and 9 respectively.

Table 7: Policy cost comparison for different season lengths and demand profiles

Cases	Average Season Length $1/q$	Demand $[\lambda_1, \lambda_2, \lambda_3]$	Optimal Average Cost g	Cost difference in %		
				Δ_1	Δ_2	Δ_3
1	10	[9,11,13]	3.303	23.35	25.56	1.90
2		[8,11,14]	3.821	12.25	18.75	6.96
3		[7,11,15]	4.912	7.01	11.83	24.18
4	20	[9,11,13]	3.306	25.03	25.52	0.85
5		[8,11,14]	3.896	12.32	18.44	5.30
6		[7,11,15]	5.541	5.73	10.39	31.40
7	100	[9,11,13]	3.289	27.58	25.89	0.11
8		[8,11,14]	3.943	13.48	19.02	1.43
9		[7,11,15]	7.970	4.24	7.67	58.92

In Table 7, policies 1 and 2 that have less production flexibility are 4-27% higher than the optimal policy depending on the seasonal conditions. Performance of policies 1 and 2 are poor when the average duration of the seasons is high. However, when the difference of the average demand among the seasons is high (Cases 3, 6, and 9), the average cost of policy 1 and 2 are closer to that of the optimal policy. Also, the performance of policy 3 is closer to the optimal policy when the average demand in each season is less than the maximum production capacity. However, where the average demand in the peak season equals the maximum production capacity, the cost of policy 3 can be 59% more than the cost of the optimal policy (Cases 3, 6, and 9). We next compare the threshold levels of the three policies in Table 8.

Comparing optimal policy and policy 1, we find that due to fixed production rate in each season, the base stock levels in policy 1 are higher than that of the optimal policy when the production rate of policy 1 is μ_{reg} (season 1), and it lies between the threshold levels of the optimal policy when the production rate of policy 1 is μ_{max} (seasons 2 and 3). However, at shorter season durations, the base-stock levels in season 1 of policy 1 may be less than that of the optimal policy due to high production rate in other seasons. Comparing optimal policy and policy 2, we find that in each of the cases, we observe that the base-stock levels of policy 2 are between the threshold levels of the optimal policy in each season. Due to high production rate μ_{max} in each season, the base-stock

Table 8: Policy comparison for different season lengths and demand profiles

Cases	Average Season Length $1/q$	Demand $[\lambda_1, \lambda_2, \lambda_3]$	Threshold Levels			
			Optimal	Policy 1	Policy 2	Policy 3
1	10	[9,11,13]	$\begin{bmatrix} 3 & 8 & 15 \\ 15 & 25 & 36 \end{bmatrix}$	[23 16 20]	[7 12 19]	$\begin{bmatrix} 2 & 6 & 20 \\ 11 & 22 & 46 \end{bmatrix}$
2		[8,11,14]	$\begin{bmatrix} 2 & 12 & 24 \\ 14 & 32 & 49 \end{bmatrix}$	[15 18 29]	[7 18 28]	$\begin{bmatrix} 1 & 6 & 44 \\ 8 & 22 & 77 \end{bmatrix}$
3		[7,11,15]	$\begin{bmatrix} 3 & 25 & 41 \\ 22 & 47 & 71 \end{bmatrix}$	[20 33 48]	[15 33 47]	$\begin{bmatrix} 0 & 6 & 600 \\ 6 & 22 & 600 \end{bmatrix}$
4	20	[9,11,13]	$\begin{bmatrix} 2 & 8 & 17 \\ 13 & 24 & 39 \end{bmatrix}$	[25 15 20]	[6 12 20]	$\begin{bmatrix} 2 & 6 & 20 \\ 11 & 22 & 46 \end{bmatrix}$
5		[8,11,14]	$\begin{bmatrix} 1 & 13 & 29 \\ 11 & 32 & 57 \end{bmatrix}$	[15 19 33]	[6 19 33]	$\begin{bmatrix} 1 & 6 & 44 \\ 8 & 22 & 77 \end{bmatrix}$
6		[7,11,15]	$\begin{bmatrix} 2 & 35 & 58 \\ 19 & 57 & 91 \end{bmatrix}$	[18 43 64]	[14 43 63]	$\begin{bmatrix} 0 & 6 & 600 \\ 6 & 22 & 600 \end{bmatrix}$
7	100	[9,11,13]	$\begin{bmatrix} 2 & 7 & 19 \\ 11 & 22 & 44 \end{bmatrix}$	[28 13 22]	[6 11 22]	$\begin{bmatrix} 2 & 6 & 20 \\ 11 & 22 & 46 \end{bmatrix}$
8		[8,11,14]	$\begin{bmatrix} 1 & 10 & 39 \\ 9 & 29 & 70 \end{bmatrix}$	[14 15 42]	[5 15 42]	$\begin{bmatrix} 1 & 6 & 44 \\ 8 & 22 & 77 \end{bmatrix}$
9		[7,11,15]	$\begin{bmatrix} 0 & 76 & 130 \\ 10 & 103 & 167 \end{bmatrix}$	[12 84 134]	[6 84 133]	$\begin{bmatrix} 0 & 6 & 600 \\ 6 & 22 & 600 \end{bmatrix}$

levels are lower than the optimal threshold levels. Finally, comparing optimal policy and policy 3, we observe that the threshold levels of policy 3 are closer to the optimal policy when season durations are longer. However, due to season interaction effects, that are not captured adequately in policy 3, the threshold levels of policy 3 are higher than that of the optimal policy in season 3 and lower than that of the optimal policy in seasons 1 and 2.

Table 9: Policy cost comparison under skewed seasonal conditions

Cases	Horizon $\sum_S \frac{1}{q_s}$	Transition Matrix Q	Demand $[\lambda_1, \lambda_2, \lambda_3]$	Optimal Cost g	Cost difference in %		
					Δ_1	Δ_2	Δ_3
1	30	$\begin{bmatrix} -1/15 & 1/15 & 0 \\ 1/10 & -2/10 & 1/10 \\ 0 & 1/5 & -1/5 \end{bmatrix}$	[9 11 17]	5.351	14.49	13.97	40.35
2	60	$\begin{bmatrix} -1/30 & 1/30 & 0 \\ 1/20 & -2/20 & 1/20 \\ 0 & 1/10 & -1/10 \end{bmatrix}$	[9 11 17]	7.092	12.20	10.53	66.27
3	300	$\begin{bmatrix} -1/150 & 1/150 & 0 \\ 1/100 & -2/100 & 1/100 \\ 0 & 1/50 & -1/50 \end{bmatrix}$	[9 11 17]	20.198	9.15	3.81	130.58

We next investigate the performance of the three policies under skewed seasonality conditions. In Table 9, we observe that the average cost of policy 3 is significantly higher than the optimal policy,

emphasizing that season interactions are important while determining the production policy especially when the seasonality in the demands are skewed. The results also show the effectiveness of policies 1 and 2 where capacity decisions are made when the season changes, independent of the current inventory status. However, unlike policy 3, both these policies optimize the inventory levels accounting for interactions between seasons, and hence seem to perform better. Further, as seen in Table 9, as the duration of each season increases, the performance of policies 1 and 2 improves while that of policy 3 degrades.

8 Model Extensions

In this section we analyze generalized versions of the model described in Section 3 and analyze the impact of relaxing these assumptions on the structure of the optimal policy. Section 8.1 analyzes a model where the season lengths have less variability and Section 8.2 analyzes a model with general cost functions.

8.1 Lower Variability in Season Durations

In Section 3, we model the season duration as a random variable with exponential distribution. In some applications, the season duration might have less variability. Such situations can be easily addressed within our modeling framework by modeling the season duration as a random variable with Erlang- r distribution. By suitably choosing the number of phases r in the Erlang- r distribution, we can reduce the coefficient of variation to $\frac{1}{r}$. Note that $r = 1$ corresponds to the exponential distribution.

If the length of each season has an Erlang- r distribution, then the state space has to be modified accordingly. Each season s is assumed to have r phases, and the mean duration of each of the r phases is $\frac{1}{r}$ times the mean duration of the season. Thus, the state space for the phases, $\mathbb{S}$, extends to $\{1, \dots, rm\}$. If we denote phases as $\hat{s} : \{1, \dots, rm\}$, then the state of the system is denoted as $(n, \hat{s})$. Further, the average demand is λ_s in each of the r phases of season s . With this enhanced state space definition, rest of the analysis can be executed as described in the previous sections.

Note that in this case, the optimal threshold levels $\{z_{\mu,\hat{s}}, z_{b,\hat{s}}\}$ are determined for each of the rm phases.

Tables 10 and 11 provide a comparison of results where the duration of each season has Exponential distribution (Table 10) and Erlang-3 distribution (Table 11). Note that when the season duration has an Exponential distribution, the manufacturer can adjust thresholds only once per season. Hence, we determine the optimal threshold levels thrice, one for each season. When each season length has an Erlang-3 distribution, the optimal threshold levels $[z_{\mu,s}, z_{b,s}]$ are set for each phase, i.e. the manufacturer can adjust the production rate thrice in each season. Thus we specify the optimal threshold levels for 9 phases corresponding to the three seasons. For cases with Exponentially distributed seasons, we specified the averaged demand for each of the three phases. In the Erlang-3 distributed seasons, there are 9 phases with identical average demands in the three phases of each season. Hence for brevity in Table 11, we specify the average demand for the first phase of each season, i.e., phases 1, 4, and 7.

Table 10: Exponentially distributed season durations

Time horizon	Demand $[\lambda_1 \ \lambda_2 \ \lambda_3]$	Optimal cost g	Optimal Policy $\begin{bmatrix} z_{\mu,1} & z_{\mu,2} & z_{\mu,3} \\ z_{b,1} & z_{b,2} & z_{b,3} \end{bmatrix}$
90	$[7 \ 9 \ 11]$	2.191	$\begin{bmatrix} 0 & 2 & 6 \\ 7 & 12 & 21 \end{bmatrix}$
300	$[7 \ 9 \ 11]$	2.187	$\begin{bmatrix} 0 & 2 & 6 \\ 6 & 11 & 21 \end{bmatrix}$

Table 11: Erlang-3 distributed season durations

Time horizon	Demand $[\lambda_1 \ \lambda_4 \ \lambda_7]$	Optimal cost g	Optimal Policy $\begin{bmatrix} z_{\mu,1} & z_{\mu,2} & z_{\mu,3} & z_{\mu,4} & z_{\mu,5} & z_{\mu,6} & z_{\mu,7} & z_{\mu,8} & z_{\mu,9} \\ z_{b,1} & z_{b,2} & z_{b,3} & z_{b,4} & z_{b,5} & z_{b,6} & z_{b,7} & z_{b,8} & z_{b,9} \end{bmatrix}$
90	$[7 \ 9 \ 11]$	2.169	$\begin{bmatrix} 0 & 0 & 0 & 2 & 2 & 2 & 6 & 6 & 6 \\ 6 & 6 & 7 & 11 & 11 & 13 & 21 & 21 & 19 \end{bmatrix}$
300	$[7 \ 9 \ 11]$	2.166	$\begin{bmatrix} 0 & 0 & 0 & 2 & 2 & 2 & 6 & 6 & 6 \\ 6 & 6 & 7 & 11 & 11 & 12 & 22 & 21 & 21 \end{bmatrix}$

As can be observed from the tables, the overall costs and levels of thresholds are fairly similar for cases with the two distributions. The difference can be attributed to the fact that when season dura-

tions have Erlang- r distribution, the manufacturer has additional flexibility to vary threshold levels multiple times in a season. As expected, this flexibility lowers the total expected cost of the system.

8.2 General Production Cost Functions

In Section 3, the production cost function is assumed to be a piecewise linear convex function of production rate with two linear segments. Due to the structure of the production cost function, the optimal policy consisted of two threshold levels $\{z_{\mu,s}, z_{b,s}\}$ in each season s . The analysis can be generalized to cases where the production cost function is a general piecewise linear convex function with $l > 2$ linear segments. Analysis along the lines of that described in Section 5.2 reveals that the optimal policy has l threshold levels for each season s . For instance, consider the production cost function

$$c(\mu) = \begin{cases} c_1\mu & 0 \leq \mu \leq \mu_1 \\ \vdots & \vdots \\ \sum_{i=1}^{l-1} c_i(\mu_i - \mu_{i-1}) + c_l(\mu - \mu_{l-1}) & \mu_{l-1} < \mu \leq \mu_l \end{cases} \quad (14)$$

where $0 = \mu_0 < \mu_1 < \dots < \mu_{l-1} < \mu_l = \mu_{max}$ and $c_1 < c_2 < \dots < c_{l-1} < c_l$, the optimal policy is

Table 12: Optimal policy as function of n

n	$\mu^*(n, s)$
$z_{1,s} \leq n$	0
$z_{2,s} \leq n < z_{1,s}$	μ_1
$\vdots$	$\vdots$
$n < z_{l,s}$	μ_l

where $z_{i,s} = \lfloor \max y^{-1}(-c_i) \rfloor$ for all $i \in \{1, \dots, l\}$.

The model and analysis also can be extended to analyze cases where the production cost functions are convex but do not necessarily have a linear or piecewise linear structure (for example $c(\mu) = e^\mu - 1$). However, in such cases, the structure of the optimal policy may not be always represented in terms of inventory threshold levels. If the inverse of the first derivative of the production cost function, $(c')^{-1}(\cdot)$, is continuous and strictly increasing, then the optimal policy can be specified in terms of $y_\alpha(n + 1, s)$.

Table 13: Optimal policy as a function of $y_\alpha(n+1, s)$

$y_\alpha(n+1, s)$	$\mu^*(n, s)$
$-c'(0) < y_\alpha(n+1, s)$	0
$-c'(\mu_{max}) < y_\alpha(n+1, s) \leq -c'(0)$	$(c')^{-1}(y_\alpha(n+1, s))$
$y_\alpha(n+1, s) < -c'(\mu_{max})$	μ_{max}

9 Concluding Remarks

In this paper, we analyzed a manufacturing system with flexible production rates and seasonal demands. The seasonal demand is modeled as a Markov modulated Poisson process (MMPP). The decision maker observes the net inventory in the system and determines production rates and target inventory levels so as to minimize the total expected costs. We modeled the system dynamics as a Markov decision process to minimize the total cost under the infinite horizon discounted expected cost and average expected cost criteria. We identified an unique optimal solution and derived an analytical expression for the optimal policy. We found that the optimal policy is a season dependent threshold policy consisting of threshold levels $z_{\mu,s}$ and $z_{b,s}$. According to the optimal policy, the production rate is μ_{max} for net inventory level, $n < z_{\mu,s}$, the production rate is μ_{reg} for $z_{\mu,s} \leq n < z_{b,s}$, and to stop the production when $z_{b,s} \leq n$. Since production cost are piecewise linear, the optimal production rates were either μ_{reg} or μ_{max} .

Next, we proved the monotonicity of the optimal policy with respect to the net inventory, n and the demand season, s . The optimal production policy is non-increasing with n for each demand season, s , and under the conditions of stochastically monotone MMPP of demands and assuming average demand increases with seasons, the optimal policy is non-decreasing with s for each net inventory level, n . We also showed that the optimal policy is non-decreasing with respect to the workload levels.

Through numerical studies, we explore the influence of various aspects of seasonality such as 1) difference of average demand across seasons 2) average length of seasons 3) skewness of seasonal demand patterns on the optimal policy and the optimal cost. We found that when the demand is highly seasonal, i.e average demands vary significantly across seasons, the optimal average cost

increases, and the optimal threshold levels differed more across the seasons. Further, in the presence of skewed seasonality in demand, leveraging both production and inventory flexibility is more valuable. The optimal average cost is also higher when the average length of each season is higher. We also observed that for a given seasonality in the demand, the optimal threshold levels differed more among the seasons at higher workload levels. Higher difference of the threshold levels within a season emphasizes the value of flexible capacity. At high workload levels, the system not only set high base-stock levels but also switched to the higher production rate at relatively higher inventory levels.

We then compared the optimal policy with three simple policies under varying seasonal conditions. Policy 1 and 2 had relatively less production flexibility compared to the optimal policy, but did incorporate the interactions across seasons while optimizing inventory. Policy 3 was a myopic policy that optimizes the threshold levels only within a season and not across the entire planning horizon. We observed that this myopic policy performs poorly, emphasizing the importance of considering interactions between seasons while determining production and inventory policies. Overall, these studies provide strong understanding of the value of capacity and inventory flexibility in production systems with seasonal demands.

Finally, we discuss generalizations of the model with respect to variability in season lengths and alternative structures of production cost functions. We analyze a system where the season duration has an Erlang- r distribution and show that the optimal policy retains a similar structure. When the production cost function has multiple linear segments, we show that the threshold levels increase and is equal to the number of linear segments in the cost function.

References

- Abhyankar, H. and Graves, S. (2001). Creating an inventory hedge for markov-modulated poisson demand: An application and model. *Manufacturing & Service Operations Management*, 3(4):306–320.
- Angelus, A. and Porteus, E. (2002). Simultaneous capacity and production management of short-

- life-cycle, produce-to-stock goods under stochastic demand. *Management Science*, 48(3):399–413.
- Beyer, D. and Sethi, S. (1997). Average cost optimality in inventory models with markovian demands. *Journal of Optimization Theory and Applications*, 92(3):497–526.
- Gayon, J., Talay-Degirmenci, I., Karaesmen, F., and Ormeci, E. (2009). Optimal pricing and production policies of a make-to-stock system with fluctuating demand. *Probability in the Engineering and Informational Sciences*, 23(2):205–230.
- Hu, J., Vakili, P., and Huang, L. (2004). Capacity and production management in a single product manufacturing systems. *Annals of Operations Research*, 125:191–204.
- Kapuscinski, R. and Tayur, S. (1998). A capacitated production-inventory model with periodic demand. *Operations Research*, 46(6):899–911.
- Keilson, J. and Kester, A. (1977). Monotone matrices and monotone markov processes. *Stochastic Processes and their Applications*, 5(3):231–241.
- Kumar, R., Lewis, M., and Topaloglu, H. (2013). Dynamic service rate control for a single-server queue with markov-modulated arrivals. *Naval Research Logistics*, 60(8):661–677.
- Lippman, S. (1975). Applying a new device in the optimization of exponential queueing systems. *Operations Research*, 23(4):687–710.
- Puterman, M. (1994). *Markov decision processes: Discrete stochastic dynamic programming*. John Wiley & Sons, Inc., Hoboken, NJ.
- Sennott, L. (1989). Average cost optimal stationary policies in infinite state markov decision processes with unbounded costs. *Operations Research*, 37(4):626–633.
- Sethi, S. and Cheng, F. (1997). Optimality of (s, S) policies in inventory models with markovian demand. *Operations Research*, 45(6):931–939.
- Shang, K. (2012). Single-stage approximations for optimal policies in serial inventory systems with non-stationary demand. *Manufacturing & Service Operations Management*, 14(3):414–422.
- Song, J. and Zipkin, P. (1993). Inventory control in a fluctuating demand environment. *Operations Research*, 41(2):351–370.

Yossi, A. and Federgruen, A. (1997). Stochastic inventory models with limited production capacity and periodically varying parameters. *Probability in the Engineering and Informational Sciences*, 11(1):107–135.

A Appendices

A.1 Proofs for Discounted Expected Costs Criterion

Since the system has unbounded cost function per transition, we cannot directly apply the Banach fixed point theorem (Puterman, 1994) to show that $\lim_{k \rightarrow \infty} v_{\alpha,k}(n, s) = v_{\alpha}(n, s)$. Instead we show that $\lim_{k \rightarrow \infty} v_{\alpha,k}(n, s) = v_{\alpha}(n, s)$ in a modified space $\mathbb{V}_{\omega}$, using Theorem 11.5.3 of Puterman (1994).

Let V be a normed linear space of bounded real-valued function with supremum norm. Define the mapping $\mathcal{L} : V \rightarrow V$ such that $v_{\alpha,k+1} = \mathcal{L}v_{\alpha,k}$ for all $v \in V$. For systems with unbounded cost functions per transition, $\mathcal{L}$ is not a single-stage contraction mapping on V and we cannot directly apply the Banach Fixed-Point Theorem. We define the new space $\mathbb{V}_{\omega}$ of real-valued functions v on $\mathbb{Z} \times \mathbb{S}$ and norm $\|\cdot\|_{\omega}$: $\|v\|_{\omega} = \sup_{n,s} \omega(n, s)^{-1} |v(n, s)|$ such that $\|v\|_{\omega} < \infty$ for all $v \in \mathbb{V}_{\omega}$, where $\omega(n, s)$ is an arbitrary positive real-valued function of $\mathbb{Z} \times \mathbb{S}$ satisfying $\inf_{n \in \mathbb{Z}, s \in \mathbb{S}} \omega(n, s) > 0$.

Next we validate three assumptions (Assumptions, 1, 2a, 2b, and 3) that ensures the discounted expected cost function of our system $v_{\alpha}^{\pi} \in \mathbb{V}_{\omega}$ for any policy $\pi \in \Pi^{MR}$, where Π^{MR} is a set of Markov randomized policies. The assumptions also ensure that the mapping $\mathcal{L}$ is a J -stage contraction mapping on $\mathbb{V}_{\omega}$, where J is a positive integer. Using the J -stage contraction property of $\mathcal{L}$, we can then show the existence of the optimal solution for infinite horizon discounted cost criterion using Theorem 11.5.3 in Puterman (1994).

Theorem 11.5.3 of Puterman (1994) states that

1. The optimality equation has a unique solution in $\mathbb{V}_{\omega}$ which equals v_{α} .
2. If a policy minimizes the right side of Equation 6, then it is a stationary deterministic optimal policy.
3. There exist an optimal stationary deterministic policy whenever the action set $\mathbb{A}$ is a compact set, and $l_{\alpha,k}(n, s, \mu)$ and $P(n', s'|n, s, \mu)$ is a continuous function of μ for $\mu \in \mathbb{A}$ and for all $(n, s) \in \mathbb{Z} \times \mathbb{S}$.
4. For each $v_{\alpha,0} \in \mathbb{V}_{\omega}$, $v_{\alpha,k} = \mathcal{L}^k v_{\alpha,0}$ converges to v_{α}

if assumptions 1, 2a, 2b, and 3 hold for our system.

Assumption 1: There exist a constant $c' < \infty$ such that

$$\sup_{\mu \in [0, \mu_{max}]} |h(n)^+ + b(n)^- + c(\mu)| \leq c' \omega(n, s)$$

$\omega(n, s)$ is any arbitrary positive real valued function of $\mathbb{Z} \times \mathbb{S}$ satisfying $\inf_{n \in \mathbb{Z}, s \in \mathbb{S}} \omega(n, s) > 0$

Choose $\omega(n, s) = (h + b)|n| + \mu_{max} c_{max}$. Assumption 1 holds for $c' = 1$.

Assumption 2a: There exist a constant $\kappa : 0 \leq \kappa < \infty$ for which

$$\sum_{n' \in \mathbb{Z}, s' \in \mathbb{S}} P(n', s'|n, s, \mu) \omega(n', s') \leq \kappa \omega(n, s) \quad \text{for } \mu \in [0, \mu_{max}] \text{ and for } n \in \mathbb{Z} \text{ and } s \in \mathbb{S} \quad (15)$$

Assumption 2b: For each r , $0 \leq r < 1$, there exists a β , $0 \leq \beta \leq 1$ and an integer J such that

$$r^J \sum_{n' \in \mathbb{Z}, s' \in \mathbb{S}} P_\pi^J(n', s' | n, s, \mu) \omega(n, s) \leq \beta \omega(n, s)$$

for all $\pi = (d_1, \dots, d_J)$ where $d_k \in D^{MD}$; $1 \leq k \leq J$

Consider the left hand side of Equation 15,

$$\begin{aligned} \sum_{n' \in \mathbb{Z}, s' \in \mathbb{S}} P(n', s' | n, s, \mu) \omega(n', s') &= \mu_{max} c_{max} + (h + b) \left[|n| \left(\frac{\nu - \lambda_s - \mu + \sum_{s' \in \mathbb{S}} Q_{s, s'}}{\nu} \right) \right. \\ &\quad \left. + |n - 1| \frac{\lambda_s}{\nu} + |n + 1| \frac{\mu}{\nu} \right] \\ &\leq \mu_{max} c_{max} + (h + b) |n| + (h + b) \\ &= \omega(n, s) + h + b \end{aligned} \tag{16}$$

Proposition 6.10.5 of Puterman (1994) states that suppose there exist a constant $L > 0$ for which

$$\sum_{n' \in \mathbb{Z}, s' \in \mathbb{S}} P(n', s' | n, s, \mu) \omega(n', s') \leq \omega(n, s) + L$$

for $a \in A_s$ and $s \in \mathbb{S}$, then Assumptions 2a and 2b hold. Assumptions 2a and 2b hold by setting $L = h + b$ and from Proposition 6.10.5 of Puterman (1994).

Assumption 3: $\nu < \infty$

Since for our system, $\lambda_{max}, \mu_{max}, \eta$ are $< \infty$, Assumption 3 holds.

Note that the conditions mentioned in part 3 of Theorem 11.5.3 can easily be validated for our system. $\mathbb{A} = [0, \mu_{max}]$ is a compact set, since $\mu_{max} < \infty$. $l_{\alpha, k}(n, s, \mu) = h(n)^+ + b(n)^- + c(\mu)$ is a continuous function of μ , and the probability function for the production term $\frac{\mu}{\nu}$ is also a continuous function of μ .

A.2 Proofs for Average Expected Cost Criterion

Theorem 1 of Sennott (1989) states that: If Equation 1 holds then

1. There exist a constant $g = \lim_{\alpha \downarrow 0} \alpha v_\alpha(n, s)$ and a vector $w(n, s) = \lim_{j \rightarrow \infty} w_{\alpha_j}(n, s) := v_{\alpha_j}(n, s) - v_{\alpha_j}(n_0, s_0)$ for a subsequence $\alpha_j \downarrow 0$ such that $(g, w(n, s))$ satisfy the Inequality 11 for $n \in \mathbb{Z}$ and $s \in \mathbb{S}$, where (n_0, s_0) is some distinguished state.
2. There exists a stationary average expected cost optimal policy $\mu^*(n, s)$ that is a limit point of a sequence of discounted expected cost optimal policies $\mu_{\alpha_j}^*(n, s)$, i.e., $\mu_{\alpha_j}^*(n, s) \rightarrow \mu^*(n, s)$ as $\alpha_j \rightarrow 0$

if assumptions 4, 5, 6, and 7 hold for our system.

Assumption 4: For each $(n, s) \in \mathbb{Z} \times \mathbb{S}$, $0 \leq l(n, s, \mu) < \infty$

Assumption 4 holds since the cost function is non-negative and is finite for each $(n, s) \in \mathbb{Z} \times \mathbb{S}$.

Assumption 5: $v_\alpha(n, s) < +\infty$ for each $(n, s) \in \mathbb{Z} \times \mathbb{S}$

Using Assumptions 1, 2a, and 2b of Appendix A.1, it can be shown that

$$v_\alpha^\pi(n, s) \leq (1 + r + \dots + (r\kappa)^{J-1}) \frac{\omega(n, s)}{1 - \beta} \text{ for all } \pi \in \Pi^{MR}$$

where Π^{MR} is a set of Markov randomized policies. Since $\omega(n, s) < \infty$ for each (n, s) , we have $v_\alpha^\pi(n, s) < \infty$. Thus, $v_\alpha(n, s) \leq v_\alpha^\pi(n, s) < \infty$.

Assumption 6: There exist a non-negative K such that $-K \leq w_\alpha(n, s)$ for $(n, s) \in \mathbb{Z} \times \mathbb{S}$.

From Proposition 3, $v_\alpha(n, s)$ attains minimum at $(n_{0,s}, s)$ for each $s \in \mathbb{S}$.

Let $(n_0, s_0) = \arg \min\{(v_\alpha(n_{0,1}, 1), \dots, (v_\alpha(n_{0,m}, m))\}$, and $w_\alpha(n, s) = v_\alpha(n, s) - v_\alpha(n_0, s_0)$, then $w_\alpha(n, s) \geq 0$ for $(n, s) \in \mathbb{Z} \times \mathbb{S}$. Then Assumption 6 holds by setting $K = 0$.

Assumption 7: There exist a non-negative number $M(n, s)$ for each $(n, s) \in \mathbb{Z} \times \mathbb{S}$ such that

1. $M(n, s) < \infty$
2. For each $(n, s) \in \mathbb{Z} \times \mathbb{S}$, $w_\alpha(n, s) \leq M(n, s)$
3. For each $(n, s) \in \mathbb{Z} \times \mathbb{S}$, there exist $\mu_{n,s} \in [0, \mu_{max}]$,

$$\sum_{n', s' \in \mathbb{Z} \times \mathbb{S}} p(n', s' | n, s, \mu_{n,s}) M(n', s') < \infty \quad (17)$$

We validate Assumption 7 using Proposition 5 of Sennott (1989). The proposition 5 of Sennott (1989) holds if there exist a stationary policy π that induces an irreducible, ergodic Markov chain and its long-run average expected cost $g^\pi < \infty$. We consider a policy π such that $\mu_{n,s} = \mu_{max}$ for $n < N_0$ and $\mu_{n,s} = 0$ for $n \geq N_0$ for all $s \in \mathbb{S}$, where N_0 is a positive integer. Then using arguments similar to Proposition 2.2 of Kumar et al. (2013), it can be shown that under the condition described in Equation 1, the Markov chain induced due to this policy is irreducible, ergodic in the state space $(-\infty, N_0] \times \mathbb{S}$, and has finite long-run expected average cost. Note that we have validated Assumption 7 under finite upper bound of the net inventory level. This condition is not restrictive for practical applications, since the inventory levels are always finite.

A.3 Proof of Proposition 1

Proposition. For each $s \in \mathbb{S}$, $v_\alpha(n, s)$ is a convex function of n .

Proof. Using induction, we first show $y_{\alpha,k}(n, s)$ is non-decreasing in n and then extend the result to $y_\alpha(n, s)$ using Lemma 1. For $k = 0$, the result holds since $v_{\alpha,0}(n, s) = 0$ for all states. For the inductive step, assume $v_{\alpha,k}(n, s)$ is a convex function of n for some k , i.e. $y_{\alpha,k}(n, s) - y_{\alpha,k}(n-1, s) \geq 0$

for $n \in \mathbb{Z}$, for $s \in \mathbb{S}$. We now prove $y_{\alpha,k+1}(n, s) - y_{\alpha,k+1}(n-1, s) \geq 0$

$$\begin{aligned}
& y_{\alpha,k+1}(n, s) - y_{\alpha,k+1}(n-1, s) \\
&= \frac{1}{\alpha + \nu} \left\{ h1_{\{n>0\}} - b1_{\{n<0\}} + \lambda_s y_{\alpha,k}(n-1, s) \right. \\
&+ \sum_{s' \neq s \in \mathbb{S}} Q_{ss'} y_{\alpha,k}(n, s') + (\nu - \lambda_s + Q_{ss}) y_{\alpha,k}(n, s) \\
&+ \min_{\mu \in \mathbb{A}} [c(\mu) + \mu y_{\alpha,k}(n+1, s)] - \min_{\mu \in \mathbb{A}} [c(\mu) + \mu y_{\alpha,k}(n, s)] \\
&- [h1_{\{n-1>0\}} - b1_{\{n-1<0\}} + \lambda_s y_{\alpha,k}(n-2, s) \\
&+ \sum_{s' \neq s \in \mathbb{S}} Q_{ss'} y_{\alpha,k}(n-1, s') + (\nu - \lambda_s + Q_{ss}) y_{\alpha,k}(n-1, s) \\
&\left. + \min_{\mu \in \mathbb{A}} [c(\mu) + \mu y_{\alpha,k}(n, s)] - \min_{\mu \in \mathbb{A}} [c(\mu) + \mu y_{\alpha,k}(n-1, s)] \right\} \quad (18)
\end{aligned}$$

Let us denote the optimal production rate $\mu_{k+1}^*(n, s)$ when the system is in state (n, s) for $(k+1)^{st}$ transition. For ease of notation, we represent $\mu_{k+1}^*(n, s)$ as μ_n^* in the remainder of this proof. Combining terms with the same co-efficients in Equation 18:

$$\begin{aligned}
& y_{\alpha,k+1}(n, s) - y_{\alpha,k+1}(n-1, s) \\
&= \frac{1}{\alpha + \nu} \left\{ h1_{\{n>0\}} - h1_{\{n-1>0\}} - b1_{\{n<0\}} + b1_{\{n-1<0\}} \right. \\
&+ \lambda_s (y_{\alpha,k}(n-1, s) - y_{\alpha,k}(n-2, s)) + (\lambda_{max} - \lambda_s) (y_{\alpha,k}(n, s) - y_{\alpha,k}(n-1, s)) \\
&+ \sum_{s' \neq s \in \mathbb{S}} Q_{ss'} (y_{\alpha,k}(n, s') - y_{\alpha,k}(n-1, s')) + (\eta + Q_{ss}) (y_{\alpha,k}(n, s) - y_{\alpha,k}(n-1, s)) \\
&+ c(\mu_n^*) + \mu_n^* y_{\alpha,k}(n+1, s) - c(\mu_{n-1}^*) + (\mu_{max} - \mu_{n-1}^*) y_{\alpha,k}(n, s) \\
&\left. - c(\mu_{n-1}^*) - \mu_{n-1}^* y_{\alpha,k}(n, s) + c(\mu_{n-2}^*) - (\mu_{max} - \mu_{n-2}^*) y_{\alpha,k}(n-1, s) \right\} \quad (19)
\end{aligned}$$

Now, replace the optimal action μ_{n-1}^* in the fourth line of the right side of Equation 19 by the suboptimal action μ_n^* ; and the optimal action μ_{n-1}^* in the fifth line of right side of Equation 19 by the suboptimal action μ_{n-2}^* . We obtain the following inequality,

$$\begin{aligned}
& y_{\alpha,k+1}(n, s) - y_{\alpha,k+1}(n-1, s) \\
&\geq \frac{1}{\alpha + \nu} \left\{ h1_{\{n>0\}} - h1_{\{n-1>0\}} - b1_{\{n<0\}} + b1_{\{n-1<0\}} \right. \\
&+ \lambda_s (y_{\alpha,k}(n-1, s) - y_{\alpha,k}(n-2, s)) + (\lambda_{max} - \lambda_s) (y_{\alpha,k}(n, s) - y_{\alpha,k}(n-1, s)) \\
&+ \sum_{s' \neq s \in \mathbb{S}} Q_{ss'} (y_{\alpha,k}(n, s') - y_{\alpha,k}(n-1, s')) + (\eta + Q_{ss}) (y_{\alpha,k}(n, s) - y_{\alpha,k}(n-1, s)) \\
&+ c(\mu_n^*) + \mu_n^* y_{\alpha,k}(n+1, s) - c(\mu_n^*) + (\mu_{max} - \mu_n^*) y_{\alpha,k}(n, s) \\
&\left. - c(\mu_{n-2}^*) - \mu_{n-2}^* y_{\alpha,k}(n, s) + c(\mu_{n-2}^*) - (\mu_{max} - \mu_{n-2}^*) y_{\alpha,k}(n-1, s) \right\} \quad (20)
\end{aligned}$$

Now, we argue that each term in the right hand side of Equation 20 is positive. It is evident that $h1_{\{n>0\}} - h1_{\{n-1>0\}}$ and $-b1_{\{n<0\}} + b1_{\{n-1<0\}}$ are both non-negative. The terms corresponding to the arrivals and phase transitions in lines 2 and 3 of right side of Equation 20 are positive due to

the induction assumption. Note that since $\eta \geq \max\{-Q_{ss}|s \in \mathbb{S}\}$, we have $\eta + Q_{ss} \geq 0$. Thus, the remaining production completion terms in lines 4 and 5 of right side of Equation 20 can be written as:

$$\mu_n^*(y_{\alpha,k}(n+1, s) - y_{\alpha,k}(n, s)) + (\mu_{\max} - \mu_{n-2}^*)(y_{\alpha,k}(n, s) - y_{\alpha,k}(n-1, s)) \quad (21)$$

Each of the two terms in Equation 21 are positive due to the induction assumption. Thus $v_{\alpha,k+1}(n, s)$ is a convex function of n . Finally, using Lemma 1, we conclude $v_\alpha(n, s)$ is a convex function of n , i.e. $y_\alpha(n, s)$ is non-decreasing in n for a given $s \in \mathbb{S}$. $\square$

A.4 Proof of Proposition 2

Proposition. For a fixed $s \in \mathbb{S}$,

1. $y_\alpha(n, s) \rightarrow \frac{-b}{\alpha}$ as $n \rightarrow -\infty$
2. $y_\alpha(n, s) \rightarrow \frac{h}{\alpha}$ as $n \rightarrow +\infty$

Proof. First we show $\frac{-b}{\alpha}$ is a lower bound and then using contradiction argument we prove $\frac{-b}{\alpha}$ is the greatest lower bound. To show $\frac{-b}{\alpha}$ is a lower bound, we first show $\frac{-b}{\alpha}$ is a lower bound for the finite horizon case using induction i.e. $y_{\alpha,k}(n, s) \geq \frac{-b}{\alpha}$ and then using Lemma 1, extend the result to infinite horizon.

Note that $y_{\alpha,0}(n, s) \equiv 0$ for $n \in \mathbb{Z}$ and $s \in \mathbb{S}$. Assume for a given k , $y_{\alpha,k}(n, s) \geq \frac{-b}{\alpha}$. We show below $y_{\alpha,k+1}(n, s) \geq \frac{-b}{\alpha}$. Replacing $(h1_{\{n \geq 0\}} - b1_{\{n < 0\}})$ by $-b$ in Equation 12 leads to the following inequality:

$$\begin{aligned} y_{\alpha,k+1}(n, s) \geq & \frac{1}{\alpha + \nu} \left[-b + \lambda_s y_{\alpha,k}(n-1, s) + \sum_{s' \in \mathbb{S}, s \neq s'} Q_{ss'} y_{\alpha,k}(n, s') + \right. \\ & (\nu - \lambda_s + Q_{ss}) y_{\alpha,k}(n, s) + \mu_{n+1}^* y_{\alpha,k}(n+1, s) + c(\mu_{n+1}^*) \\ & \left. - \mu_n^* y_{\alpha,k}(n, s) - c(\mu_n^*) \right] \text{ for } n \in \mathbb{Z} \text{ and for } s \in \mathbb{S} \end{aligned}$$

Replacing the optimal μ_n^* by the suboptimal μ_{n+1}^* and substituting $y_{\alpha,k}(n, s)$ by its lower bound $\frac{-b}{\alpha}$, we obtain:

$$\begin{aligned} y_{\alpha,k+1}(n, s) \geq & \frac{1}{\alpha + \nu} \left[-b + \lambda_s \left(\frac{-b}{\alpha} \right) + \sum_{s' \in \mathbb{S}, s \neq s'} Q_{ss'} \left(\frac{-b}{\alpha} \right) + (\nu - \lambda_s + Q_{ss} - \mu_{n+1}^*) \left(\frac{-b}{\alpha} \right) \right. \\ & \left. - c(\mu_{n+1}^*) + \mu_{n+1}^* \left(\frac{-b}{\alpha} \right) + c(\mu_{n+1}^*) \right] \\ = & \frac{-b}{\alpha} \quad \text{for } n \in \mathbb{Z} \text{ and for } s \in \mathbb{S} \end{aligned}$$

Thus, $\frac{-b}{\alpha}$ is a lower bound of $y_{\alpha,k}(n, s)$ for $k \geq 0$.

Next, we show by contradiction that $\frac{-b}{\alpha}$ is the greatest lower bound of $y_{\alpha,k}(n, s)$. Let $(\frac{-b'}{\alpha})$ be the greatest lower bound of $y_{\alpha,k}(n, s)$ for $k \geq 0$ such that $b' \neq b$ and $b - b' = \delta > 0$. Since $y_{\alpha,k}(n, s)$ is non-decreasing in n , it converges asymptotically to $\frac{-b'}{\alpha}$ as $n \rightarrow \infty$. For a given $\frac{\delta}{\nu} > 0$, $\exists N$: for

$s \in \mathbb{S}$ and for $n \leq N$, we have $y_{\alpha,k}(n, s) < \frac{-b'}{\alpha} + \frac{\delta}{\nu}$. Replacing $y_{\alpha,k}(n, s)$ by $\frac{-b'}{\alpha} + \frac{\delta}{\nu}$ in Equation 12, we obtain:

$$\begin{aligned} y_{\alpha,k+1}(n, s) &< \frac{1}{\alpha + \nu} \left[-b + \lambda_s \left(\frac{-b}{\alpha} + \frac{\delta}{\nu} \right) + (\nu - \lambda_s + Q_{ss} - \mu_n^*) \left(\frac{-b}{\alpha} + \frac{\delta}{\nu} \right) \right. \\ &\quad \left. + \sum_{s' \in \mathbb{S}, s \neq s'} Q_{ss'} \left(\frac{-b}{\alpha} + \frac{\delta}{\nu} \right) - c(\mu_n^*) + \mu_{n+1}^* \left(\frac{-b}{\alpha} + \frac{\delta}{\nu} \right) + c(\mu_{n+1}^*) \right] \end{aligned} \quad (22)$$

Replacing the optimal μ_{n+1}^* by the suboptimal μ_n^* , we obtain:

$$\begin{aligned} y_{\alpha,k+1}(n, s) &< \frac{1}{\alpha + \nu} \left[-b + \lambda_s \left(\frac{-b}{\alpha} + \frac{\delta}{\nu} \right) + \sum_{s' \in \mathbb{S}, s \neq s'} Q_{ss'} \left(\frac{-b}{\alpha} + \frac{\delta}{\nu} \right) \right. \\ &\quad \left. + (\nu - \lambda_s + Q_{ss} - \mu_n^*) \left(\frac{-b}{\alpha} + \frac{\delta}{\nu} \right) - c(\mu_n^*) + \mu_n^* \left(\frac{-b}{\alpha} + \frac{\delta}{\nu} \right) + c(\mu_n^*) \right] \\ &= \frac{-b'}{\alpha} \end{aligned} \quad (23)$$

which is a contradiction since $\frac{-b'}{\alpha}$ was assumed to be the greatest lower bound. Thus $\frac{-b}{\alpha}$ is the greatest lower bound of $y_{\alpha,k}(n, s)$ for $k \geq 0$.

Finally, using Lemma 1, we extend the result to the infinite horizon case, i.e. $\frac{-b}{\alpha}$ is the greatest lower bound of $y_{\alpha}(n, s)$ for $k \geq 0$. Part 2 of Proposition 2 is proved similarly. $\square$

A.5 Proof of Proposition 3

Proposition: For each $s \in \mathbb{S}$, there exist a finite $n_{o,s} \in \mathbb{Z}$ such that $v_{\alpha}(n, s)$ is decreasing in n at the fixed s , for all $-\infty < n < n_{o,s}$ and increasing in n for all $n_{o,s} \leq n < \infty$.

Proof. We prove the result by contradiction that $-\infty < n_{o,s} < \infty$. If we assume that $n_{o,s} = \infty$ then as $n \rightarrow +\infty$, $y_{\alpha}(n, s) \rightarrow 0$ since $y_{\alpha}(n, s)$ is non-decreasing in n for each s and $0 < \frac{h}{\alpha}$. This is a contradiction since from Part 1 of Proposition 2, $\frac{h}{\alpha}$ is the least upper bound of $y_{\alpha}(n, s)$. Hence, $n_{o,s} < \infty$. Similarly, we can show that $n_{o,s} > -\infty$ by observing that $0 > -\frac{b}{\alpha}$ and as $n \rightarrow -\infty$, $y_{\alpha}(n, s) \rightarrow \frac{-b}{\alpha}$. $\square$

A.6 Proof of Theorem 3

Theorem. If the phase transition process is stochastically monotone and $\lambda_1 < \lambda_2 < \dots < \lambda_m$, then:

1. There exists a discounted expected cost optimal policy $\mu_{\alpha}^*(n, s)$ which is non-decreasing in s for each $n \in \mathbb{Z}$
2. There exists an average expected cost optimal policy $\mu^*(n, s)$ which is non-decreasing in s for each $n \in \mathbb{Z}$

Proof. Using induction, we first show the result for finite horizon case i.e., $y_{\alpha,k}(n+1, s)$ is non-increasing with respect to s for each $n \in \mathbb{Z}$. For $k = 0$, $y_{\alpha,0}(n, s) \equiv 0$. Assume for a given k , $y_{\alpha,k}(n+1, s) - y_{\alpha,k}(n+1, s-1) \leq 0$. We now prove the inequality for $(k+1)^{st}$ transition. For ease

of notation, we represent $\mu_{\alpha,k+1}(n, s)$ to $\mu_{n,s}$ in the remainder of this proof.

$$\begin{aligned}
& y_{\alpha,k+1}(n+1, s) - y_{\alpha,k+1}(n+1, s-1) \\
&= \frac{1}{\alpha + \nu} \left\{ \lambda_s y_{\alpha,k}(n, s) - \lambda_{s-1} y_{\alpha,k}(n, s-1) \right. \\
&+ \sum_{s' \neq s \in \mathbb{S}} Q_{ss'} y_{\alpha,k}(n+1, s') - \sum_{s' \neq s-1 \in \mathbb{S}} Q_{s-1,s'} y_{\alpha,k}(n+1, s') \\
&+ (\lambda_{max} - \lambda_s + \eta + Q_{ss}) y_{\alpha,k}(n+1, s) \\
&- (\lambda_{max} - \lambda_{s-1} + \eta + Q_{s-1,s-1}) y_{\alpha,k}(n+1, s-1) \\
&+ \mu_{max} (y_{\alpha,k}(n+1, s) - y_{\alpha,k}(n+1, s-1)) \\
&+ \min_{\mu \in \mathbb{A}} [c(\mu) + \mu y_{\alpha,k}(n+2, s)] - \min_{\mu \in \mathbb{A}} [c(\mu) + \mu y_{\alpha,k}(n+1, s)] \\
&\left. - \min_{\mu \in \mathbb{A}} [c(\mu) + \mu y_{\alpha,k}(n+2, s-1)] + \min_{\mu \in \mathbb{A}} [c(\mu) + \mu y_{\alpha,k}(n+1, s-1)] \right\}
\end{aligned}$$

Adding and subtracting $\lambda_s(y_{\alpha,k}(n, s-1) + y_{\alpha,k}(n+1, s-1))$ and then combining terms with same coefficients yield:

$$\begin{aligned}
& y_{\alpha,k+1}(n+1, s) - y_{\alpha,k+1}(n+1, s-1) \\
&= \frac{1}{\alpha + \nu} \left\{ (\lambda_s - \lambda_{s-1}) (y_{\alpha,k}(n, s-1) - y_{\alpha,k}(n+1, s-1)) \right. \\
&+ \lambda_s (y_{\alpha,k}(n, s) - y_{\alpha,k}(n, s-1)) + (\lambda_{max} - \lambda_s) (y_{\alpha,k}(n+1, s) - y_{\alpha,k}(n+1, s-1)) \\
&+ \sum_{s', s \in \mathbb{S}} \bar{Q}_{ss'} y_{\alpha,k}(n+1, s') - \sum_{s', s-1 \in \mathbb{S}} \bar{Q}_{s-1,s'} y_{\alpha,k}(n+1, s') \\
&+ c(\mu_{n+1,s}^*) + \mu_{n+1,s}^* y_{\alpha,k}(n+2, s) - c(\mu_{n,s}^*) + (\mu_{max} - \mu_{n,s}^*) y_{\alpha,k}(n+1, s) \\
&- c(\mu_{n+1,s-1}^*) - \mu_{n+1,s-1}^* y_{\alpha,k}(n+2, s-1) \\
&\left. + c(\mu_{n,s-1}^*) - (\mu_{max} - \mu_{n,s-1}^*) y_{\alpha,k}(n+1, s-1) \right\} \tag{24}
\end{aligned}$$

where $\bar{Q} = \eta I + Q$. We now show that each term in the right side of Equation 24 is non-positive. The first term is non-positive due to convexity of $v_{\alpha,k}(n, s-1)$. The next two terms in lines 1 and 2 of right side of Equation 24 are non-positive due to the induction assumption. We next use the stochastically monotone matrix property to show that $\sum_{s,s' \in \mathbb{S}} \bar{Q}_{ss'} y_{\alpha,k}(n+1, s') - \sum_{s-1,s' \in \mathbb{S}} \bar{Q}_{s-1,s'} y_{\alpha,k}(n+1, s') \leq 0$. Since the transition probability matrix of the MMPP is stochastically monotone, from Theorem 1.1 of Keilson and Kester (1977), $\bar{Q}x$ is non-increasing for all non-increasing vectors x . $y_{\alpha,k}(n, s)$ is non-decreasing in s by induction assumption. Hence, $\sum_{s,s' \in \mathbb{S}} \bar{Q}_{ss'} y_{\alpha,k}(n+1, s')$ is nondecreasing in s , i.e., $\sum_{s,s' \in \mathbb{S}} \bar{Q}_{ss'} y_{\alpha,k}(n+1, s') - \sum_{s-1,s' \in \mathbb{S}} \bar{Q}_{s-1,s'} y_{\alpha,k}(n+1, s') \leq 0$.

Now consider the production completion terms in lines 4, 5 and 6 of right side of Equation 24 which are highlighted again in the left side of Equation 25. Replacing the optimal $\mu_{n+1,s}^*$ in the first line of Equation 25 by the suboptimal $\mu_{n+1,s-1}^*$ and the optimal $\mu_{n,s-1}^*$ in the second line of Equation 25 by the suboptimal $\mu_{n,s}^*$ yields the following inequality

$$\begin{aligned}
& c(\mu_{n+1,s}^*) + \mu_{n+1,s}^* y_{\alpha,k}(n+2, s) - c(\mu_{n,s}^*) + (\mu_{max} - \mu_{n,s}^*) y_{\alpha,k}(n+1, s) \\
& - c(\mu_{n+1,s-1}^*) - \mu_{n+1,s-1}^* y_{\alpha,k}(n+2, s-1) + c(\mu_{n,s-1}^*) - (\mu_{max} - \mu_{n,s-1}^*) y_{\alpha,k}(n+1, s-1) \\
& \leq \\
& c(\mu_{n+1,s-1}^*) + \mu_{n+1,s-1}^* y_{\alpha,k}(n+2, s) - c(\mu_{n,s}^*) + (\mu_{max} - \mu_{n,s}^*) y_{\alpha,k}(n+1, s) \\
& - c(\mu_{n+1,s-1}^*) - \mu_{n+1,s-1}^* y_{\alpha,k}(n+2, s-1) + c(\mu_{n,s}^*) - (\mu_{max} - \mu_{n,s}^*) y_{\alpha,k}(n+1, s-1)
\end{aligned} \tag{25}$$

Rearranging the terms on the right side of Equation 25 yields,

$$\mu_{n+1,s-1}^* \left(y_{\alpha,k}(n+2, s) - y_{\alpha,k}(n+2, s-1) \right) + \left(\mu_{max} - \mu_{n,s}^* \right) \left(y_{\alpha,k}(n+1, s) - y_{\alpha,k}(n+1, s-1) \right) \tag{26}$$

Both the terms in Equation 26 are non-positive due to the induction assumption. Thus, $y_{\alpha,k}(n, s)$ is non-increasing in s for $k \geq 0$. Using Lemma 1, $y_{\alpha}(n, s)$ is non-increasing in s for $n \in \mathbb{Z}$ and for $s \in \mathbb{S}$. Part 2 of the Theorem 3 is proved along the same lines as part 3 of Theorem 2 $\square$

A.7 Proof of Theorem 4

Theorem. For $\epsilon \geq 0$, if $\mu_{\alpha,\epsilon}^*(n, s)$ is a discounted cost optimal policy for a system with demand $\Lambda_\epsilon = [\lambda_1 + \epsilon, \dots, \lambda_m + \epsilon]$ then $\mu_{\alpha,\epsilon}^*(n, s)$ is non-decreasing with ϵ as long as $\mu_{max} > \sum_{s \in \mathbb{S}} \hat{\pi}_{s,\epsilon}(\lambda_s + \epsilon)$.

Proof. We first show the result for the finite horizon using induction and then using Lemma 1, extend the result to the infinite horizon case. For $k = 0$, $y_{\alpha,0}(n, s) = y_{\alpha,\epsilon,0}(n, s) = 0$. Assume $y_{\alpha,k}(n, s) - y_{\alpha,\epsilon,k}(n, s) \geq 0$. We now show $y_{\alpha,k+1}(n, s) - y_{\alpha,\epsilon,k+1}(n, s) \geq 0$. From Equation 12, we have

$$\begin{aligned}
& y_{\alpha,k+1}(n, s) - y_{\alpha,\epsilon,k+1}(n, s) \\
& = \frac{1}{\alpha + \nu} \left\{ h1_{\{n>0\}} - b1_{\{n<0\}} + \lambda_s y_{\alpha,k}(n-1, s) + \sum_{s' \neq s \in \mathbb{S}} Q_{ss'} y_{\alpha,k}(n, s') \right. \\
& \quad + (\nu - \lambda_s + Q_{ss}) y_{\alpha,k}(n, s) + \min_{\mu \in \mathbb{A}} \left[c(\mu) + \mu y_{\alpha,k}(n+1, s) \right] - \min_{\mu \in \mathbb{A}} \left[c(\mu) + \mu y_{\alpha,k}(n, s) \right] \\
& \quad - \left[h1_{\{n>0\}} - b1_{\{n<0\}} + (\lambda_s + \epsilon) y_{\alpha,\epsilon,k}(n-1, s) + \sum_{s' \neq s \in \mathbb{S}} Q_{ss'} y_{\alpha,\epsilon,k}(n, s') \right. \\
& \quad \left. + (\nu - \lambda_s - \epsilon + Q_{ss}) y_{\alpha,\epsilon,k}(n, s) + \min_{\mu \in \mathbb{A}} \left[c(\mu) + \mu y_{\alpha,\epsilon,k}(n+1, s) \right] \right. \\
& \quad \left. \left. - \min_{\mu \in \mathbb{A}} \left[c(\mu) + \mu y_{\alpha,\epsilon,k}(n, s) \right] \right] \right\}
\end{aligned}$$

Combining terms with same coefficients yields:

$$\begin{aligned}
& y_{\alpha,k+1}(n, s) - y_{\alpha,\epsilon,k+1}(n, s) \\
&= \frac{1}{\alpha + \nu} \left\{ \lambda_s \left(y_{\alpha,k}(n-1, s) - y_{\alpha,\epsilon,k}(n-1, s) \right) + (\lambda_{max} - \lambda_s) \left(y_{\alpha,k}(n, s) - y_{\alpha,\epsilon,k}(n, s') \right) \right. \\
&+ \sum_{s', s \in \mathbb{S}} \bar{Q}_{ss'} \left(y_{\alpha,k}(n, s') - y_{\alpha,\epsilon,k}(n, s') \right) + \epsilon \left(y_{\alpha,\epsilon,k}(n, s) - y_{\alpha,\epsilon,k}(n-1, s) \right) \\
&+ c(\mu_{n,s}^*) + \mu_{n,s}^* y_{\alpha,k}(n+1, s) - c(\mu_{n-1,s}^*) + (\mu_{max} - \mu_{n-1,s}^*) y_{\alpha,k}(n, s) \\
&\left. - c(\mu_{n,s,\epsilon}^*) - \mu_{n,s,\epsilon}^* y_{\alpha,\epsilon,k}(n+1, s) + c(\mu_{n-1,s,\epsilon}^*) - (\mu_{max} - \mu_{n-1,s,\epsilon}^*) y_{\alpha,\epsilon,k}(n, s) \right\} \quad (27)
\end{aligned}$$

We now show that each combined term of the right side of Equation 27 is non-negative. The first three terms are non-negative due to induction assumption. Next, $\epsilon \left(y_{\alpha,\epsilon,k}(n, s) - y_{\alpha,\epsilon,k}(n-1, s) \right)$ is non-negative due to the convexity property of $v_{\alpha,\epsilon,k}(n, s)$. Finally, the production completion terms in lines 3 and 4 of Equation 27 are re-written as the left side of Equation 28. Replacing the optimal $\mu_{n-1,s}^*$ in the first line of Equation 28 by the suboptimal $\mu_{n-1,s,\epsilon}^*$ and the optimal $\mu_{n,s,\epsilon}^*$ in the second line of Equation 28 by the suboptimal $\mu_{n,s}^*$, we obtain the following inequality:

$$\begin{aligned}
& c(\mu_{n,s}^*) + \mu_{n,s}^* y_{\alpha,k}(n+1, s) - c(\mu_{n-1,s}^*) + (\mu_{max} - \mu_{n-1,s}^*) y_{\alpha,k}(n, s) \\
& - c(\mu_{n,s,\epsilon}^*) - \mu_{n,s,\epsilon}^* y_{\alpha,\epsilon,k}(n+1, s) + c(\mu_{n-1,s,\epsilon}^*) - (\mu_{max} - \mu_{n-1,s,\epsilon}^*) y_{\alpha,\epsilon,k}(n, s) \\
& \geq \\
& c(\mu_{n,s}^*) + \mu_{n,s}^* y_{\alpha,k}(n+1, s) - c(\mu_{n-1,s,\epsilon}^*) + (\mu_{max} - \mu_{n-1,s,\epsilon}^*) y_{\alpha,k}(n, s) \\
& - c(\mu_{n,s}^*) - \mu_{n,s}^* y_{\alpha,\epsilon,k}(n+1, s) + c(\mu_{n-1,s,\epsilon}^*) - (\mu_{max} - \mu_{n-1,s,\epsilon}^*) y_{\alpha,\epsilon,k}(n, s) \quad (28)
\end{aligned}$$

Rearranging the terms on the right side of the Equation 28 yield,

$$\mu_{n,s}^* \left(y_{\alpha,k}(n+1, s) - y_{\alpha,\epsilon,k}(n+1, s) \right) + (\mu_{max} - \mu_{n-1,s,\epsilon}^*) \left(y_{\alpha,k}(n, s) - y_{\alpha,\epsilon,k}(n, s) \right) \quad (29)$$

The last two terms in Equation 29 are non-negative due to the induction assumption. Thus, $y_{\alpha,k+1}(n, s) - y_{\alpha,\epsilon,k+1}(n, s) \geq 0$ for $n \in \mathbb{Z}$ and for $s \in \mathbb{S}$.

Finally, by Lemma 1, we extend the results to infinite horizon to obtain $y_{\alpha}(n, s) - y_{\alpha,\epsilon}(n, s) \geq 0$. $\square$

This result can also be generalized to the case where the increase in the average demand is not equal among the seasons. i.e. if $\mu'(n, s)$ is the optimal policy for a system with average demand $\Lambda' = [\lambda_1 + \epsilon_1, \dots, \lambda_m + \epsilon_m]$ where $\epsilon_i \geq 0$ for $i \in [1, \dots, m]$ then $\mu'_{\alpha}(n, s) \geq \mu_{\alpha}(n, s)$ for all $n \in \mathbb{Z}$ and $s \in \mathbb{S}$ as long as the stability condition Equation 1 holds in both cases.